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Advanced Macroeconomics I

Problem Notes and Selected Solutions for Romer 5ed

COURSE

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PREFACE

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CHAPTER 1

PROBLEM 1.1 *Basic properties of growth rates.*

Use the fact that the growth rate of a variable equals the time derivative of its log to show:

- (a) The growth rate of the product of two variables equals the sum of their growth rates. That is, if $Z(t) = X(t)Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)} \quad (1)$$

- (b) The growth rate of the ratio of two variables equals the difference of their growth rates. That is, if $Z(t) = X(t)/Y(t)$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)} \quad (2)$$

- (c) If $Z(t) = aX(t)^\alpha$, then

$$\frac{\dot{Z}(t)}{Z(t)} = \alpha \frac{\dot{X}(t)}{X(t)} \quad (3)$$

SOLUTION

For any positive differentiable variable $X(t)$, define its instantaneous growth rate by

$$g_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{d \ln X(t)}{dt}. \quad (4)$$

- (a) If $Z(t) = X(t)Y(t)$, then

$$\begin{aligned} \ln Z(t) &= \ln X(t) + \ln Y(t), \\ \frac{d \ln Z(t)}{dt} &= \frac{d \ln X(t)}{dt} + \frac{d \ln Y(t)}{dt}. \end{aligned}$$

Therefore,

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)}}.$$

(b) If $Z(t) = X(t)/Y(t)$, then

$$\begin{aligned}\ln Z(t) &= \ln X(t) - \ln Y(t), \\ \frac{d \ln Z(t)}{dt} &= \frac{d \ln X(t)}{dt} - \frac{d \ln Y(t)}{dt}.\end{aligned}$$

Hence,

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)}}.$$

(c) Assume $a > 0$ and let $Z(t) = aX(t)^\alpha$, where a and α are constants. Taking logs and then differentiating gives

$$\begin{aligned}\ln Z(t) &= \ln a + \alpha \ln X(t), \\ \frac{d \ln Z(t)}{dt} &= \frac{d}{dt} [\ln a + \alpha \ln X(t)] \\ &= 0 + \alpha \frac{d \ln X(t)}{dt}.\end{aligned}$$

Using the growth-rate identity in (4) for both $Z(t)$ and $X(t)$, we obtain

$$\boxed{\frac{\dot{Z}(t)}{Z(t)} = \alpha \frac{\dot{X}(t)}{X(t)}}.$$

Thus a time-invariant multiplicative constant changes the level of a variable, but not its growth rate.

NOTE WHY LOGARITHMS WORK. Because instantaneous growth in continuous time is naturally expressed in exponential form, $\lim_{m \rightarrow \infty} (1 + g/m)^{mt} = e^{gt}$, the rules above reflect a useful duality: logarithms turn products into sums, while exponentials turn sums into products.

NOTE FOUR GROWTH-RATE RULES. For products, add the growth rates. For ratios, subtract them. For powers, multiply the growth rate by the exponent. Multiplying a variable by a constant does not change its growth rate.

NOTE SUMS AND DIFFERENCES. Sums and differences also have growth-rate formulas, but there is no analogous unweighted rule. For example, the growth rate of $X + Y$ is a level-weighted average of the growth rates of X and Y .

PROBLEM 1.2 Growth rates and log levels.

Suppose that the growth rate of some variable, X , is constant and equal to $a > 0$ from time 0 to time t_1 ; drops to 0 at time t_1 ; rises gradually from 0 to a from time t_1 to time t_2 ; and is constant and equal to a after time t_2 .

(a) Sketch a graph of the growth rate of X as a function of time.

(b) Sketch a graph of $\ln X$ as a function of time.

SOLUTION

Let

$$\gamma_X(t) \equiv \frac{\dot{X}(t)}{X(t)} = \frac{d \ln X(t)}{dt}. \quad (1)$$

(a) The question does not specify the exact shape of the gradual increase between t_1 and t_2 . For a concrete sketch, suppose it is linear. Then

$$\gamma_X(t) = \begin{cases} a, & 0 \leq t < t_1, \\ \frac{a(t - t_1)}{t_2 - t_1}, & t_1 \leq t \leq t_2, \\ a, & t > t_2. \end{cases} \quad (2)$$

Thus the growth rate is initially horizontal at a , jumps down to zero at t_1 , rises to a , and is horizontal again after t_2 .

(b) Since the slope of $\ln X(t)$ equals $\gamma_X(t)$,

$$\ln X(t) - \ln X(0) = \int_0^t \gamma_X(\tau) d\tau. \quad (3)$$

Substituting the linear path in (2) into (3) gives

$$\ln X(t) = \begin{cases} \ln X(0) + at, & 0 \leq t \leq t_1, \\ \ln X(0) + at_1 + \frac{a(t - t_1)^2}{2(t_2 - t_1)}, & t_1 \leq t \leq t_2, \\ \ln X(0) + at_1 + \frac{a(t_2 - t_1)}{2} + a(t - t_2), & t \geq t_2. \end{cases} \quad (4)$$

Hence $\ln X(t)$ is linear with slope a before t_1 . It remains continuous at t_1 , but its slope drops to zero. Between t_1 and t_2 it is increasing and convex, and after t_2 it is again linear with slope a .

The two paths derived above are collected in the figure below. The left panel shows the assumed growth-rate path, and the right panel shows its implication for the level of $\ln X(t)$.

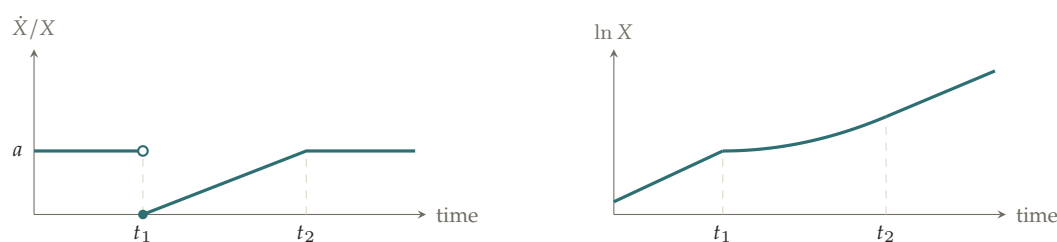


Figure 1.1. The growth rate of X and the corresponding path of $\ln X$. The straight portions of the right-hand panel have slope a .

NOTE READING A LOG-LEVEL GRAPH. A jump in a growth rate changes the slope of the log-level path, not its level. Thus $\ln X(t)$ has a kink at t_1 , not a jump or an inflection point.

PROBLEM 1.3 *Comparative statics in the Solow diagram.*

Describe how, if at all, each of the following developments affects the break-even and actual investment lines in our basic diagram for the Solow model:

- (a) The rate of depreciation falls.
- (b) The rate of technological progress rises.
- (c) The production function is Cobb–Douglas, $f(k) = k^\alpha$, and capital's share, α , rises.
- (d) Workers exert more effort, so that output per unit of effective labor for a given value of capital per unit of effective labor is higher than before.

SOLUTION

The law of motion for capital per unit of effective labor is

$$\dot{k} = sf(k) - (n + g + \delta)k. \quad (1)$$

The balanced-growth-path value k^* is determined by

$$sf(k^*) = (n + g + \delta)k^*. \quad (2)$$

- (a) By (1), a fall in δ reduces $n + g + \delta$, so the break-even investment line becomes flatter. The actual investment curve $sf(k)$ is unchanged. At the old k^* , actual investment now exceeds break-even investment, and therefore

$$k_{\text{new}}^* > k_{\text{old}}^*.$$

- (b) A rise in g increases $n + g + \delta$, so the break-even investment line becomes steeper. The actual investment curve is unchanged. At the old k^* , break-even investment now exceeds actual investment, and therefore

$$k_{\text{new}}^* < k_{\text{old}}^*.$$

- (c) With $f(k) = k^\alpha$, actual investment is sk^α , while the break-even line is unaffected. For a fixed k ,

$$\frac{\partial(sk^\alpha)}{\partial\alpha} = sk^\alpha \ln k. \quad (3)$$

Thus an increase in α lowers actual investment when $0 < k < 1$, leaves it unchanged at $k = 1$, and raises it when $k > 1$. The actual investment curve therefore rotates around $k = 1$; it does not shift uniformly upward or downward.

Solving the steady-state condition (2) for the Cobb–Douglas case gives

$$k^* = \left(\frac{s}{n + g + \delta} \right)^{1/(1-\alpha)}. \quad (4)$$

Taking logs and differentiating with respect to α gives

$$\frac{\partial \ln k^*}{\partial \alpha} = \frac{\ln(s/(n + g + \delta))}{(1 - \alpha)^2}. \quad (5)$$

Consequently,

$$\begin{aligned} s > n + g + \delta &\implies k^* \text{ rises,} \\ s < n + g + \delta &\implies k^* \text{ falls,} \\ s = n + g + \delta &\implies k^* = 1 \text{ and is unchanged.} \end{aligned}$$

Without additional information, the effect on k^* is ambiguous.

- (d) Let the higher effort level raise output at every relevant k , so that the new production function satisfies $\tilde{f}(k) > f(k)$. Actual investment shifts upward from $s f(k)$ to $s \tilde{f}(k)$, while the break-even line is unchanged. Hence

$$k_{\text{new}}^* > k_{\text{old}}^*.$$

The four comparative-static cases are collected in the figure below. In each panel, the vertical guides mark the old and new balanced-growth-path values of capital per unit of effective labor.

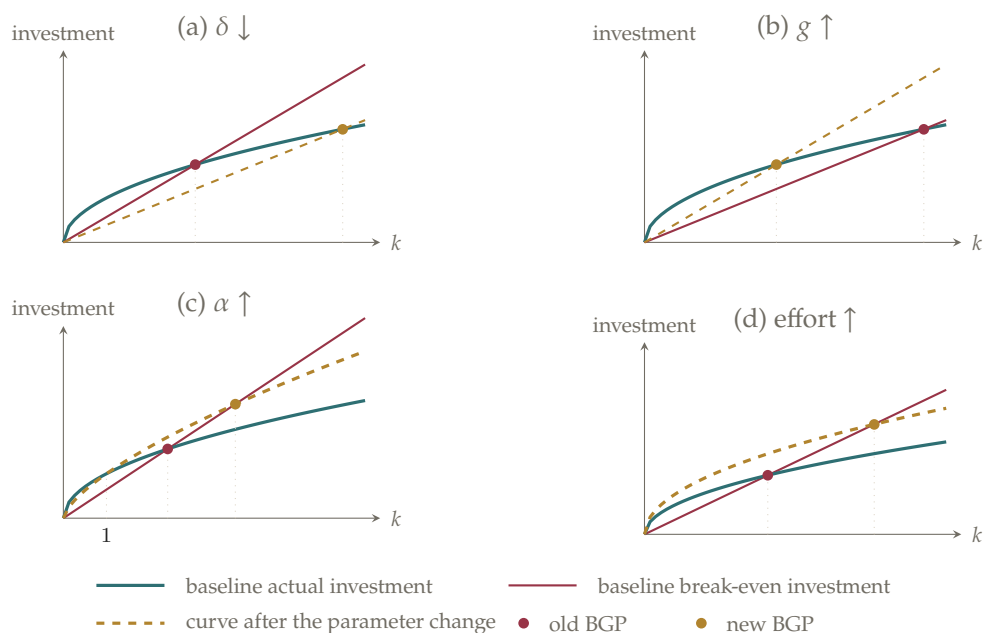


Figure 1.2. Comparative statics in the Solow diagram. Vertical guides locate the old and new balanced-growth-path capital stocks.

NOTE DIAGRAM STRATEGY. First identify which curve changes. Parameters n , g , and δ rotate the break-even line; productivity or effort changes the actual-investment curve.

NOTE CHANGING AN EXPONENT. In part (c), a higher α rotates rather than uniformly shifts sk^α : it lowers the curve when $0 < k < 1$, leaves it unchanged at $k = 1$, and raises it when $k > 1$.

PROBLEM 1.4 *A one-time increase in the labor force.*

Consider an economy with technological progress but without population growth that is on its balanced growth path. Now suppose there is a one-time jump in the number of workers.

- (a) At the time of the jump, does output per unit of effective labor rise, fall, or stay the same? Why?
- (b) After the initial change (if any) in output per unit of effective labor when the new workers appear, is there any further change in output per unit of effective labor? If so, does it rise or fall? Why?
- (c) Once the economy has again reached a balanced growth path, is output per unit of effective labor higher, lower, or the same as it was before the new workers appeared? Why?

SOLUTION

Let the one-time increase in labor occur at t_0 . Immediately before the shock, the economy is on a balanced growth path, and

$$k(t_0^-) = \frac{K(t_0^-)}{A(t_0^-)L(t_0^-)} = k^*. \quad (1)$$

Suppose $L(t_0^+) = mL(t_0^-)$, where $m > 1$. Capital K and technology A cannot jump at that instant, so $K(t_0^+) = K(t_0^-)$ and $A(t_0^+) = A(t_0^-)$.

- (a) Immediately after the increase in workers,

$$\begin{aligned} k(t_0^+) &= \frac{K(t_0^+)}{A(t_0^+)L(t_0^+)} \\ &= \frac{K(t_0^-)}{A(t_0^-)mL(t_0^-)} \\ &= \frac{k(t_0^-)}{m} = \frac{k^*}{m} < k^*. \end{aligned} \quad (2)$$

Since $y = f(k)$ and $f'(k) > 0$,

$$y(t_0^+) = f(k(t_0^+)) < f(k(t_0^-)) = y(t_0^-) = y^*. \quad (3)$$

Thus output per unit of effective labor falls immediately: the existing capital stock is spread across more effective labor.

(b) Because there is no population growth, the post-shock law of motion remains

$$\dot{k} = s f(k) - (g + \delta)k. \quad (4)$$

The one-time change in the level of L does not alter either curve in the Solow diagram. Since (2) shows that $k(t_0^+) < k^*$,

$$s f(k(t_0^+)) > (g + \delta)k(t_0^+),$$

so $\dot{k} > 0$. Capital per unit of effective labor rises toward k^* , and $y = f(k)$ rises with it.

(c) The balanced-growth-path condition is still

$$s f(k^*) = (g + \delta)k^*. \quad (5)$$

Since none of the parameters in this equation has changed, the long-run values of k^* and $y^* = f(k^*)$ are unchanged. Therefore,

$$y_{\text{new}}^* = y_{\text{old}}^*.$$

The transition after the one-time labor increase is summarized in the figure below. A single long arrow represents the instantaneous jump, whereas successive arrowheads represent the economy's gradual movement under the unchanged law of motion.

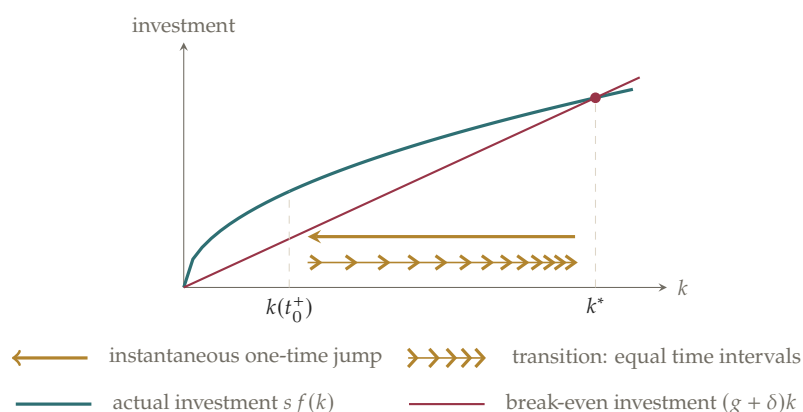


Figure 1.3. A one-time increase in L moves the economy left of k^* , but does not shift either curve. The shrinking gaps between successive rightward arrowheads indicate that the transition slows as k approaches k^* .

NOTE AGGREGATION. Do not confuse total output Y with output per unit of effective labor y . More workers generally raise Y on impact even though $y = Y/(AL)$ falls.

NOTE LEVEL VERSUS GROWTH SHOCK. This is a one-time level shock to L , not a permanent change in population growth. It moves the economy away from its old balanced growth path without shifting either curve in the Solow diagram.

PROBLEM 1.5 *The golden rule of the Cobb–Douglas Solow model.*

Suppose that the production function is Cobb–Douglas.

- (a) Find expressions for k^* , y^* , and c^* as functions of the parameters of the model, s , n , δ , g , and α .
- (b) What is the golden-rule value of k ?
- (c) What saving rate is needed to yield the golden-rule capital stock?

SOLUTION

Let the intensive-form production function be

$$f(k) = k^\alpha, \quad 0 < \alpha < 1. \quad (1)$$

The law of motion for capital per unit of effective labor is

$$\dot{k} = sk^\alpha - (n + g + \delta)k. \quad (2)$$

- (a) On the balanced growth path, $\dot{k} = 0$. Thus

$$s(k^*)^\alpha = (n + g + \delta)k^*. \quad (3)$$

For the positive steady state, divide by k^* and rearrange:

$$(k^*)^{1-\alpha} = \frac{s}{n + g + \delta}.$$

Therefore,

$$k^* = \left(\frac{s}{n + g + \delta} \right)^{1/(1-\alpha)}. \quad (4)$$

Since $y = f(k) = k^\alpha$,

$$y^* = \left(\frac{s}{n + g + \delta} \right)^{\alpha/(1-\alpha)}. \quad (5)$$

Finally, $c = (1 - s)y$, and hence

$$c^* = (1 - s) \left(\frac{s}{n + g + \delta} \right)^{\alpha/(1-\alpha)}. \quad (6)$$

- (b) On a balanced growth path, investment equals break-even investment. Consumption per unit of effective labor can therefore be written as

$$c(k) = f(k) - (n + g + \delta)k = k^\alpha - (n + g + \delta)k. \quad (7)$$

The golden-rule capital stock maximizes consumption in (7). The first-order condition is

$$\frac{dc(k)}{dk} = \alpha k^{\alpha-1} - (n + g + \delta) = 0,$$

or equivalently

$$f'(k_{GR}) = n + g + \delta. \quad (8)$$

Solving for k_{GR} gives

$$k_{GR} = \left(\frac{\alpha}{n + g + \delta} \right)^{1/(1-\alpha)}. \quad (9)$$

Moreover,

$$\frac{d^2c(k)}{dk^2} = \alpha(\alpha - 1)k^{\alpha-2} < 0,$$

so this critical point is the unique maximum.

- (c) A saving rate s_{GR} supports the golden-rule capital stock if

$$s_{GR}f(k_{GR}) = (n + g + \delta)k_{GR}. \quad (10)$$

Since $f(k) = k^\alpha$,

$$s_{GR} = (n + g + \delta)k_{GR}^{1-\alpha}.$$

For Cobb–Douglas production, (8) implies $k_{GR}^{1-\alpha} = \alpha/(n + g + \delta)$. Hence

$$s_{GR} = \alpha. \quad (11)$$

Under competitive factor pricing, α is also capital's gross share of output.

NOTE STEADY STATE VERSUS GOLDEN RULE. The steady-state condition $sf(k^*) = (n + g + \delta)k^*$ determines the capital stock generated by a given saving rate. The golden-rule condition $f'(k_{GR}) = n + g + \delta$ instead identifies the capital stock that maximizes steady-state consumption.

NOTE COBB–DOUGLAS SHORTCUT. The result $s_{GR} = \alpha$ is specific to Cobb–Douglas production. Under competitive factor pricing, it says that the golden-rule saving rate equals capital's gross share of output.

PROBLEM 1.6 *A permanent fall in population growth.*

Consider a Solow economy that is on its balanced growth path. Assume for simplicity that there is no technological progress. Now suppose that the rate of population growth falls.

- (a) What happens to the balanced-growth-path values of capital per worker, output per worker, and consumption per worker? Sketch the paths of these variables as the economy moves to its new balanced growth path.
- (b) Describe the effect of the fall in population growth on the path of output (that is, total output, not output per worker).

SOLUTION

Because there is no technological progress, A is constant. We can normalize it to one and write

$$k \equiv \frac{K}{L}, \quad y \equiv \frac{Y}{L} = f(k), \quad c \equiv \frac{C}{L} = (1 - s)f(k). \quad (1)$$

Suppose the population growth rate falls permanently at t_0 from n_{old} to $n_{new} < n_{old}$.

- (a) Capital per worker evolves according to

$$\dot{k} = sf(k) - (n + \delta)k. \quad (2)$$

Before the change, the economy is on its balanced growth path, so

$$sf(k_{old}^*) = (n_{old} + \delta)k_{old}^*. \quad (3)$$

Equation (2) shows that the fall in n rotates the break-even investment line downward from $(n_{old} + \delta)k$ to $(n_{new} + \delta)k$. It does not change $sf(k)$. Therefore, at the old steady state,

$$\begin{aligned} \dot{k}(t_0^+) &= sf(k_{old}^*) - (n_{new} + \delta)k_{old}^* \\ &= (n_{old} - n_{new})k_{old}^* > 0. \end{aligned} \quad (4)$$

Thus capital per worker rises monotonically until it reaches the new balanced-

growth-path value:

$$\boxed{k_{\text{new}}^* > k_{\text{old}}^*}.$$

A change in a growth rate does not create an instantaneous level jump in K or L . Hence

$$k(t_0^+) = k(t_0^-) = k_{\text{old}}^*. \quad (5)$$

After t_0 , k rises gradually. Since $f'(k) > 0$ and s is unchanged, output and consumption per worker also rise gradually:

$$\boxed{y_{\text{new}}^* > y_{\text{old}}^*, \quad c_{\text{new}}^* > c_{\text{old}}^*}.$$

(b) Total output satisfies $\dot{Y} = Ly$, so

$$\frac{\dot{Y}}{Y} = \frac{\dot{L}}{L} + \frac{\dot{y}}{y} = n + \frac{\dot{y}}{y}. \quad (6)$$

On the initial balanced growth path, $\dot{y} = 0$, so (6) becomes

$$\left. \frac{\dot{Y}}{Y} \right|_{\text{old BGP}} = n_{\text{old}}. \quad (7)$$

On the new balanced growth path, $\dot{y} = 0$ again, and therefore

$$\left. \frac{\dot{Y}}{Y} \right|_{\text{new BGP}} = n_{\text{new}}. \quad (8)$$

There is no jump in Y at t_0 . Immediately after the change, K is still growing at its old balanced-growth-path rate n_{old} , while labor grows at n_{new} . Growth accounting therefore gives

$$\frac{\dot{Y}(t_0^+)}{Y(t_0^+)} = \alpha_K(k_{\text{old}}^*) n_{\text{old}} + [1 - \alpha_K(k_{\text{old}}^*)] n_{\text{new}}, \quad (9)$$

which lies strictly between n_{new} and n_{old} . During the transition, output continues to grow, but its growth rate gradually approaches n_{new} . Hence total output falls increasingly below the path it would have followed if population growth had remained at n_{old} , even though output per worker is higher.

The steady-state comparison and the transition paths from part (a) are collected in the first figure below. The second figure separates the actual path of total output in part (b) from the no-change counterfactual.

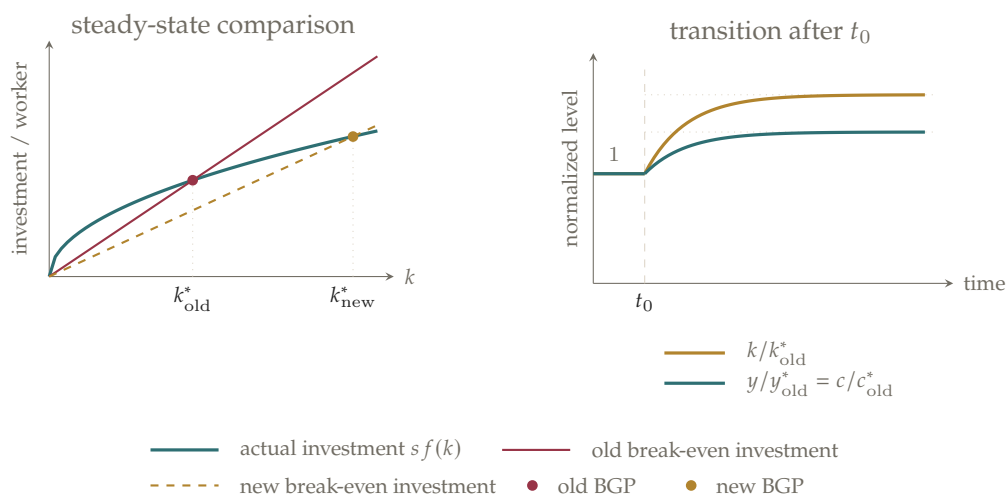


Figure 1.4. A permanent fall in population growth raises capital, output, and consumption per worker. These variables do not jump at t_0 ; they rise gradually toward their new balanced-growth-path levels.

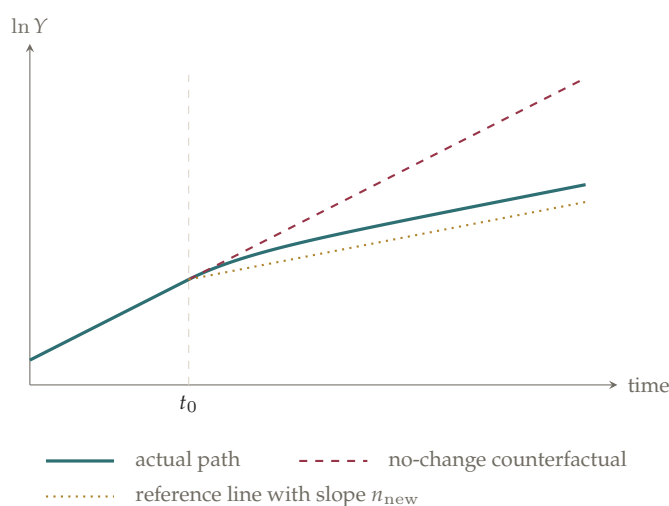


Figure 1.5. Total output is continuous at t_0 . Its growth rate falls below n_{old} immediately and converges gradually to n_{new} .

NOTE CONTINUITY. A fall in the *growth rate* of labor does not reduce the level of labor on impact. Per-worker variables are therefore continuous at t_0 , unlike the one-time level increase in L studied in Problem 1.4.

PROBLEM 1.7 *The population-growth elasticity of steady-state output.*

Find the elasticity of output per unit of effective labor on the balanced growth path, y^* , with respect to the rate of population growth, n . If $\alpha_K(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^* ?

SOLUTION

The balanced-growth-path conditions are

$$y^* = f(k^*), \quad (1)$$

$$s f(k^*) = (n + g + \delta)k^*. \quad (2)$$

Differentiate the steady-state condition (2) with respect to n :

$$s f'(k^*) \frac{\partial k^*}{\partial n} = k^* + (n + g + \delta) \frac{\partial k^*}{\partial n}.$$

Collecting the terms that contain $\partial k^* / \partial n$ gives

$$[s f'(k^*) - (n + g + \delta)] \frac{\partial k^*}{\partial n} = k^*,$$

and hence

$$\frac{\partial k^*}{\partial n} = \frac{k^*}{s f'(k^*) - (n + g + \delta)}.$$

Use the steady-state condition to write

$$s = \frac{(n + g + \delta)k^*}{f(k^*)}.$$

Then

$$\begin{aligned} s f'(k^*) &= (n + g + \delta) \frac{k^* f'(k^*)}{f(k^*)} \\ &= (n + g + \delta) \alpha_K(k^*), \end{aligned}$$

where

$$\alpha_K(k^*) \equiv \frac{k^* f'(k^*)}{f(k^*)}. \quad (3)$$

Therefore,

$$\frac{\partial k^*}{\partial n} = - \frac{k^*}{(n + g + \delta) [1 - \alpha_K(k^*)]} < 0. \quad (4)$$

Since $y^* = f(k^*)$, the elasticity of y^* with respect to n is

$$\begin{aligned}\varepsilon_{y^*,n} &\equiv \frac{n}{y^*} \frac{\partial y^*}{\partial n} \\ &= \frac{n}{f(k^*)} f'(k^*) \frac{\partial k^*}{\partial n} \\ &= \boxed{-\frac{n}{n+g+\delta} \frac{\alpha_K(k^*)}{1-\alpha_K(k^*)}}.\end{aligned}\tag{5}$$

For a rough calculation, evaluate (5) at the midpoint $\bar{n} = 1.5\%$:

$$\begin{aligned}\varepsilon_{y^*,n} &\approx -\frac{0.015}{0.015+0.02+0.03} \frac{1/3}{1-1/3} \\ &= -0.115 \approx -0.12.\end{aligned}$$

The fall from 2% to 1% is a 50% fall relative to the initial rate. The conventional textbook-style elasticity calculation therefore gives

$$\frac{\Delta y^*}{y^*} \approx (-0.12)(-0.50) = 0.06.$$

Thus the standard rough answer is

$$\boxed{y^* \text{ rises by about } 6\%}.$$

There is a small numerical subtlety because this is a large change in n . If $\alpha_K = 1/3$ is treated as constant over the interval, a base-consistent log-change calculation gives

$$\begin{aligned}\Delta \ln y^* &\approx -\frac{1/3}{1-1/3} \ln\left(\frac{0.01+0.02+0.03}{0.02+0.02+0.03}\right) \\ &= \frac{1}{2} \ln\left(\frac{0.07}{0.06}\right) \approx 0.077,\end{aligned}$$

which corresponds to an increase of about 8%. The difference reflects the crudeness of applying a local elasticity to a 50% change.

NOTE LOCAL VERSUS FINITE CHANGES. For grading against the standard solution, the expected back-of-the-envelope answer is about 6%. A consistent finite-change calculation is closer to 8% because halving n is not a small change.

NOTE ECONOMIC INTUITION. Population growth enters through $n + g + \delta$, so a change in n alters only part of the total break-even-investment rate. Plausible differences in population growth therefore have only modest level effects on y^* .

PROBLEM 1.8 *A permanent increase in the investment rate.*

Suppose that investment as a fraction of output in the United States rises permanently from 0.15 to 0.18. Assume that capital's share is $1/3$.

- By about how much does output eventually rise relative to what it would have been without the rise in investment?
- By about how much does consumption rise relative to what it would have been without the rise in investment?
- What is the immediate effect of the rise in investment on consumption? About how long does it take for consumption to return to what it would have been without the rise in investment?

SOLUTION

Let $s_{\text{old}} = 0.15$ and $s_{\text{new}} = 0.18$. The proportional increase in the investment rate is

$$\frac{s_{\text{new}} - s_{\text{old}}}{s_{\text{old}}} = \frac{0.03}{0.15} = 0.20.$$

Thus the saving rate rises by 20%, not by 3%; the latter number is the change in percentage points.

- From equation (1.28) in the text, the balanced-growth-path elasticity of output with respect to the saving rate is

$$\varepsilon_{y^*,s} \equiv \frac{s}{y^*} \frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1 - \alpha_K(k^*)}. \quad (1)$$

With $\alpha_K = 1/3$, (1) becomes

$$\varepsilon_{y^*,s} = \frac{1/3}{1 - 1/3} = \frac{1}{2}.$$

Hence the elasticity approximation gives

$$\frac{\Delta y^*}{y^*} \approx \frac{1}{2}(0.20) = 0.10.$$

Therefore,

Long-run output is about 10% higher.

If a Cobb–Douglas production function with capital share $1/3$ is used to calculate

the finite change exactly, then

$$\frac{y_{\text{new}}^*}{y_{\text{old}}^*} = \left(\frac{s_{\text{new}}}{s_{\text{old}}} \right)^{(1/3)/(2/3)} = \sqrt{1.2} \approx 1.095,$$

so the exact increase is 9.5%, which confirms the approximation.

(b) Balanced-growth-path consumption per unit of effective labor is

$$c^* = (1 - s)y^*. \quad (2)$$

Therefore, using (2),

$$\begin{aligned} \frac{c_{\text{new}}^*}{c_{\text{old}}^*} &= \frac{1 - s_{\text{new}}}{1 - s_{\text{old}}} \frac{y_{\text{new}}^*}{y_{\text{old}}^*} \\ &= \frac{0.82}{0.85} \sqrt{1.2} \\ &\approx 1.057. \end{aligned}$$

Thus

Long-run consumption is about 6% higher.

Consumption rises less than output because the economy devotes a larger fraction of its higher output to investment.

(c) At the instant the saving rate rises, K , A , L , and hence y , cannot jump. Consumption does jump because the consumed fraction of output falls immediately:

$$\begin{aligned} \frac{c(t_0^+)}{c(t_0^-)} &= \frac{(1 - s_{\text{new}})y(t_0)}{(1 - s_{\text{old}})y(t_0)} \\ &= \frac{0.82}{0.85} \approx 0.965. \end{aligned}$$

Hence

Consumption falls immediately by about 3.5%.

To estimate the recovery time, use the text's benchmark $n + g + \delta \approx 0.06$. The local convergence rate is

$$\begin{aligned} \lambda &= [1 - \alpha_K(k^*)] (n + g + \delta) \\ &= \left(1 - \frac{1}{3} \right) (0.06) = 0.04. \end{aligned} \quad (3)$$

Normalize consumption on the old path to one, and define

$$x_0 \equiv \frac{c(t_0^+)}{c_{\text{old}}^*} = \frac{0.82}{0.85} \approx 0.965,$$

$$x_{\text{new}}^* \equiv \frac{c_{\text{new}}^*}{c_{\text{old}}^*} \approx 1.057.$$

The linearized transition is

$$x(t) - x_{\text{new}}^* \approx e^{-\lambda(t-t_0)} (x_0 - x_{\text{new}}^*). \quad (4)$$

Consumption has returned to its old path when $x(t) = 1$. Substituting this condition into (4) gives

$$e^{-\lambda(t-t_0)} = \frac{x_{\text{new}}^* - 1}{x_{\text{new}}^* - x_0}$$

$$\approx \frac{1.057 - 1}{1.057 - 0.965} \approx 0.62.$$

Taking logs gives

$$t - t_0 \approx -\frac{\ln(0.62)}{0.04}$$

$$\approx 12 \text{ years}.$$

Therefore,

Consumption returns to its old path after roughly 12 years.

The impact effect and the gradual recovery of consumption in part (c) are summarized in the figure below. The horizontal benchmark is the consumption path that would have prevailed without the increase in investment.

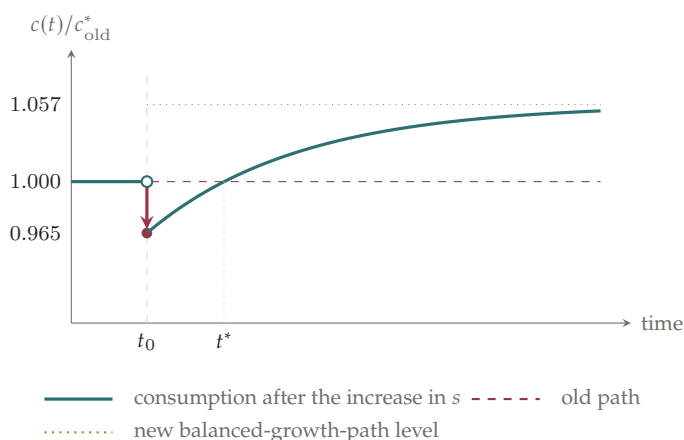


Figure 1.6. Consumption falls on impact, crosses its old path at $t^* \approx t_0 + 12$ years, and then approaches a level about 6% above the old path.

PROBLEM 1.9 *Factor payments in the Solow model.*

Assume that both labor and capital are paid their marginal products. Let w denote $\partial F(K, AL) / \partial L$ and r denote $[\partial F(K, AL) / \partial K] - \delta$.

- (a) Show that the marginal product of labor, w , is $A [f(k) - k f'(k)]$.
- (b) Show that if both capital and labor are paid their marginal products, constant returns to scale imply that the total amount paid to the factors of production equals total net output. That is, show that under constant returns, $wL + rK = F(K, AL) - \delta K$.
- (c) The return to capital (r) is roughly constant over time, as are the shares of output going to capital and to labor. Does a Solow economy on a balanced growth path exhibit these properties? What are the growth rates of w and r on a balanced growth path?
- (d) Suppose the economy begins with a level of k less than k^* . As k moves toward k^* , is w growing at a rate greater than, less than, or equal to its growth rate on the balanced growth path? What about r ?

SOLUTION

Write the constant-returns production function in intensive form:

$$Y = F(K, AL) = AL f(k), \quad k \equiv \frac{K}{AL}. \quad (1)$$

- (a) The wage is the marginal product of labor, holding K and A fixed. First,

$$\frac{\partial k}{\partial L} = \frac{\partial}{\partial L} \left(\frac{K}{AL} \right) = -\frac{K}{AL^2} = -\frac{k}{L}.$$

Apply the product and chain rules to $Y = AL f(k)$:

$$\begin{aligned} w \equiv \frac{\partial Y}{\partial L} &= A f(k) + AL f'(k) \frac{\partial k}{\partial L} \\ &= A f(k) + AL f'(k) \left(-\frac{k}{L} \right) \\ &= \boxed{A [f(k) - k f'(k)]}. \end{aligned} \quad (2)$$

- (b) First derive the marginal product of capital. Since

$$\frac{\partial k}{\partial K} = \frac{1}{AL},$$

we have

$$\frac{\partial Y}{\partial K} = AL f'(k) \frac{1}{AL} = f'(k). \quad (3)$$

Therefore the net return to capital is

$$r = f'(k) - \delta. \quad (4)$$

Total labor income is

$$\begin{aligned} wL &= AL [f(k) - kf'(k)] \\ &= ALf(k) - ALkf'(k) \\ &= Y - Kf'(k), \end{aligned}$$

where $ALk = K$. Total net capital income is

$$rK = Kf'(k) - \delta K.$$

Adding the two payments gives

$$\begin{aligned} wL + rK &= Y - Kf'(k) + Kf'(k) - \delta K \\ &= \boxed{F(K, AL) - \delta K}. \end{aligned} \quad (5)$$

Equation (5) is Euler's theorem for a constant-returns production function, with depreciation deducted from capital's payment.

(c) On a balanced growth path, $k = k^*$ is constant. Hence

$$r = f'(k^*) - \delta \quad (6)$$

is constant, so

$$\boxed{\frac{\dot{r}}{r} = 0}. \quad (7)$$

Evaluating the wage formula (2) at k^* gives

$$w = A [f(k^*) - k^* f'(k^*)]. \quad (8)$$

The expression in brackets is constant on the balanced growth path, while A grows at rate g . Therefore,

$$\boxed{\frac{\dot{w}}{w} = g}. \quad (9)$$

The factor shares are constant as well. For example, r is constant, and both K and

Y grow at $n + g$, so

$$\frac{d}{dt} \ln \left(\frac{rK}{Y} \right) = 0 + (n + g) - (n + g) = 0. \quad (10)$$

Likewise, w grows at g , L grows at n , and Y grows at $n + g$, so wL/Y is constant. The shares in net output are also constant because $Y - \delta K$, wL , and rK all grow at $n + g$ on the balanced growth path.

(d) Let

$$h(k) \equiv f(k) - kf'(k), \quad (11)$$

so that $w = Ah(k)$. Its derivative is

$$h'(k) = f'(k) - [f'(k) + kf''(k)] = -kf''(k).$$

Thus

$$\begin{aligned} \frac{\dot{w}}{w} &= \frac{\dot{A}}{A} + \frac{h'(k)\dot{k}}{h(k)} \\ &= g - \frac{kf''(k)\dot{k}}{f(k) - kf'(k)}. \end{aligned} \quad (12)$$

When $k < k^*$, stability implies $\dot{k} > 0$. With diminishing marginal returns, $f''(k) < 0$, and labor's gross marginal product satisfies $f(k) - kf'(k) > 0$. Therefore,

$$\boxed{\frac{\dot{w}}{w} > g}.$$

The wage grows faster than on the balanced growth path because workers benefit from both technological progress and capital deepening.

For the net return to capital,

$$r = f'(k) - \delta, \quad \dot{r} = f''(k)\dot{k} < 0. \quad (13)$$

If $r > 0$, its proportional growth rate is

$$\frac{\dot{r}}{r} = \frac{f''(k)\dot{k}}{f'(k) - \delta} < 0. \quad (14)$$

Hence

$$\boxed{r \text{ falls during the transition, so its growth rate is below 0.}}$$

It approaches the constant balanced-growth-path return from above.

NOTE NET VERSUS GROSS RETURN. The problem defines $r = f'(k) - \delta$, the *net* return. Therefore $\dot{k}/k = f''(k)\dot{k}/[f'(k) - \delta]$; using $f'(k)$ in the denominator would instead be the growth rate of the gross marginal product of capital.

PROBLEM 1.10 *Returns, growth, and the capital-output ratio.*

(Piketty, 2014; Piketty and Zucman, 2014; Rognlie, 2015) This problem asks you to use a Solow-style model to investigate some ideas that have been discussed in the context of Thomas Piketty's recent work. Consider an economy described by the assumptions of the Solow model, except that factors are paid their marginal products (as in the previous problem), and all labor income is consumed and all other income is saved. Thus, $C(t) = L(t) [\partial Y(t) / \partial L(t)]$.

- Show that the properties of the production function and our assumptions about the behavior of L and A imply that the capital-output ratio, K/Y , is rising if and only if the growth rate of K is greater than $n + g$ —that is, if and only if k is rising.
- Assume that the initial conditions are such that $\partial Y / \partial K$ at $t = 0$ is strictly greater than $n + g + \delta$. Describe the qualitative behavior of the capital-output ratio over time. (For example, does it grow or fall without bound? Gradually approach some constant level from above or below? Something else?) Explain your reasoning.
- Many popular summaries of Piketty's work describe his thesis as: Since the return to capital exceeds the growth rate of the economy, the capital-output ratio tends to grow without bound. By the assumptions in part (b), this economy starts in a situation where the return to capital exceeds the economy's growth rate. If you found in (b) that K/Y grows without bound, explain intuitively whether the driving force of this unbounded growth is that the return to capital exceeds the economy's growth rate. Alternatively, if you found in (b) that K/Y does not grow without bound, explain intuitively what is wrong with the statement that the return to capital exceeding the economy's growth rate tends to cause K/Y to grow without bound.
- Suppose $F(\cdot)$ is Cobb–Douglas and that the initial situation is as in part (b). Describe the qualitative behavior over time of the share of net capital income (that is, $K(t) [\partial Y(t) / \partial K(t) - \delta]$) in net output (that is, $Y(t) - \delta K(t)$). Explain your reasoning. Is the common statement that an excess of the return to capital over the economy's growth rate causes capital's share to rise over time correct in this case?

SOLUTION

Let $F_K \equiv \partial F(K, AL) / \partial K$, and define capital's gross output elasticity by

$$\alpha_K(t) \equiv \frac{F_K(t)K(t)}{Y(t)}. \quad (1)$$

Because the production function has constant returns to scale, labor income plus gross

capital income exhausts output. Since all labor income is consumed,

$$Y - C = F_K K. \quad (2)$$

The capital accumulation equation is therefore

$$\begin{aligned} \dot{K} &= Y - C - \delta K \\ &= (F_K - \delta) K. \end{aligned} \quad (3)$$

Thus the growth rate of capital equals the net return to capital:

$$\frac{\dot{K}}{K} = F_K - \delta. \quad (4)$$

(a) Differentiate $Y = F(K, AL)$ with respect to time:

$$\dot{Y} = F_K \dot{K} + F_{AL} (\dot{AL}).$$

Divide by Y , multiply and divide the first term by K , and multiply and divide the second term by AL :

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{Y} \frac{\dot{K}}{K} + \frac{F_{AL} (AL)}{Y} \frac{\dot{AL}}{AL}.$$

Constant returns imply

$$\frac{F_K K}{Y} + \frac{F_{AL} (AL)}{Y} = 1, \quad (5)$$

and AL grows at $n + g$. Hence

$$\frac{\dot{Y}}{Y} = \alpha_K \frac{\dot{K}}{K} + (1 - \alpha_K)(n + g). \quad (6)$$

The growth rate of the capital-output ratio is therefore

$$\begin{aligned} \frac{d}{dt} \ln \left(\frac{K}{Y} \right) &= \frac{\dot{K}}{K} - \frac{\dot{Y}}{Y} \\ &= (1 - \alpha_K) \left[\frac{\dot{K}}{K} - (n + g) \right]. \end{aligned} \quad (7)$$

With $0 < \alpha_K < 1$, this expression is positive if and only if

$$\frac{\dot{K}}{K} > n + g.$$

Finally, since $k = K/(AL)$,

$$\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - (n + g). \quad (8)$$

Thus

$\frac{K}{Y} \text{ rises} \iff \frac{\dot{K}}{K} > n + g \iff \dot{k} > 0.$
--

(b) In intensive form, $F_K = f'(k)$. Using the accumulation equation (3),

$$\begin{aligned}\frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - (n + g) \\ &= f'(k) - (n + g + \delta).\end{aligned}\tag{9}$$

The assumption in the question is

$$f'(k(0)) > n + g + \delta,$$

so $\dot{k}(0) > 0$. Under the usual Solow assumptions, $f''(k) < 0$, so the marginal product $f'(k)$ falls as k rises. Let $\bar{k}$ denote the unique value satisfying

$$f'(\bar{k}) = n + g + \delta.\tag{10}$$

Then

$$\begin{aligned}k < \bar{k} &\implies \dot{k} > 0, \\ k > \bar{k} &\implies \dot{k} < 0.\end{aligned}$$

Since the initial condition implies $k(0) < \bar{k}$, capital per unit of effective labor rises monotonically toward $\bar{k}$.

Moreover,

$$\frac{K}{Y} = \frac{k}{f(k)},\tag{11}$$

and

$$\frac{d \ln[k/f(k)]}{d \ln k} = 1 - \frac{k f'(k)}{f(k)} = 1 - \alpha_K(k) > 0.\tag{12}$$

Therefore K/Y rises whenever k rises and converges to the finite value $\bar{k}/f(\bar{k})$. Hence

$\frac{K}{Y} \text{ approaches a finite constant from below.}$

(c) At the start of the transition,

$$r \equiv F_K - \delta > n + g.\tag{13}$$

Since $\dot{K}/K = r$, capital initially grows faster than effective labor, and K/Y rises. But the return r is not fixed: as k rises, diminishing marginal returns reduce $F_K = f'(k)$, and hence reduce r . The process stops when

$$r = n + g,\tag{14}$$

equivalently when $f'(k) = n + g + \delta$. At that point k and K/Y are constant.

Thus the statement that r exceeding the economy's growth rate causes K/Y to grow

without bound incorrectly treats the return to capital as exogenous. In this model,

capital accumulation lowers r until the excess return disappears.

(d) With Cobb–Douglas production,

$$Y = K^\alpha (AL)^{1-\alpha}, \quad 0 < \alpha < 1, \quad (15)$$

so

$$F_K = \alpha \frac{Y}{K}. \quad (16)$$

Net capital income is

$$K(F_K - \delta) = \alpha Y - \delta K, \quad (17)$$

while net output is $Y - \delta K$. Let

$$\beta \equiv \frac{K}{Y}. \quad (18)$$

The share of net capital income in net output is

$$\begin{aligned} \theta_K^{\text{net}} &\equiv \frac{\alpha Y - \delta K}{Y - \delta K} \\ &= \frac{\alpha - \delta\beta}{1 - \delta\beta}. \end{aligned} \quad (19)$$

Differentiate with respect to β :

$$\begin{aligned} \frac{d\theta_K^{\text{net}}}{d\beta} &= \frac{-\delta(1 - \delta\beta) - (\alpha - \delta\beta)(-\delta)}{(1 - \delta\beta)^2} \\ &= \frac{\delta(\alpha - 1)}{(1 - \delta\beta)^2} < 0. \end{aligned} \quad (20)$$

Equations (11) and (12) show that $\beta = K/Y$ rises monotonically. Therefore,

θ_K^{net} falls monotonically toward a constant.

Thus, in the Cobb–Douglas case, an initial excess of the return to capital over the economy’s growth rate is associated with a *fall*, not a rise, in capital’s share of net output.

The qualitative paths established in parts (b)–(d) are collected in the figure below. Each horizontal guide marks the corresponding long-run value.

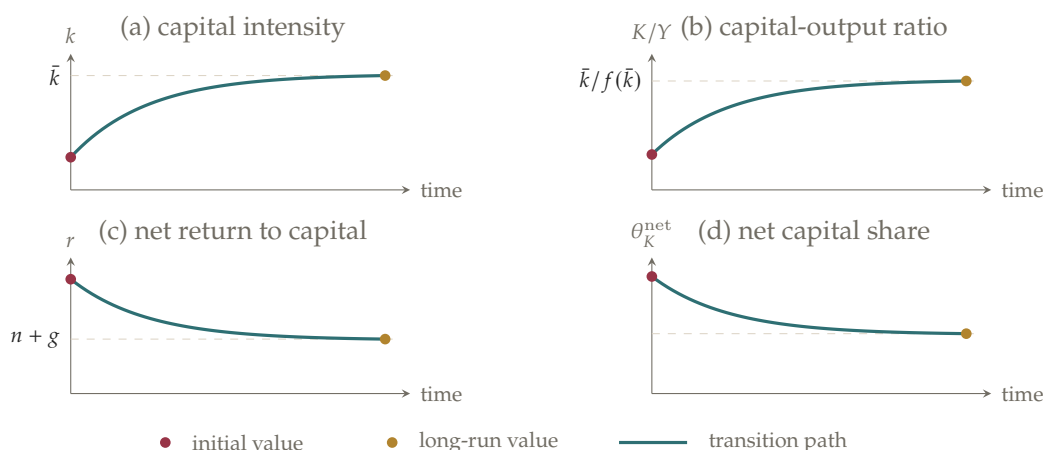


Figure 1.7. With $f'(k(0)) > n + g + \delta$, capital intensity and the capital-output ratio rise, while the net return falls to $n + g$. Under Cobb–Douglas production, capital’s share of net output also falls.

NOTE TRANSITION, NOT A PARAMETER RESTRICTION. The comparison $r > g_Y$ is not a permanent parameter restriction here. It is a transitional inequality: capital deepening lowers the marginal product of capital until $r = n + g$ on the balanced growth path.

PROBLEM 1.11 *Net capital shares and the elasticity of substitution.*

Consider Problem 1.10. Suppose there is a marginal increase in K .

- Derive an expression (in terms of K/Y , δ , the marginal product of capital F_K , and the elasticity of substitution between capital and effective labor in the gross production function $F(\cdot)$) that determines whether a marginal increase in K increases, reduces, or has no effect on the share of net capital income in net output.
- Suppose the capital-output ratio is 3, $\delta = 3\%$, and the rate of return on capital $(F_K - \delta) = 5\%$. How large must the elasticity of substitution in the gross production function be for the share of net capital income in net output to rise when K rises?

SOLUTION

Let

$$\theta_K^{\text{net}} \equiv \frac{K(F_K - \delta)}{Y - \delta K} = \frac{F_K - \delta}{Y/K - \delta} \quad (1)$$

denote capital’s share of net output. We maintain the usual assumptions that $F_K > \delta$, $Y/K > \delta$, and $F_{KK} < 0$.

- Along a constant-returns-to-scale production function, the elasticity of substitution

between capital and effective labor can be written as

$$\sigma = -\frac{\partial \ln(K/Y)}{\partial \ln F_K} = \frac{\partial \ln(Y/K)}{\partial \ln F_K}. \quad (2)$$

Hence

$$\frac{\partial(Y/K)}{\partial F_K} = \sigma \frac{Y/K}{F_K}. \quad (3)$$

Differentiate (1) with respect to F_K :

$$\begin{aligned} \frac{\partial \theta_K^{\text{net}}}{\partial F_K} &= \frac{1}{Y/K - \delta} - \frac{F_K - \delta}{(Y/K - \delta)^2} \frac{\partial(Y/K)}{\partial F_K} \\ &= \frac{(Y/K - \delta) - \sigma(F_K - \delta)(Y/K)/F_K}{(Y/K - \delta)^2}. \end{aligned} \quad (4)$$

Since F_K falls when K rises, the sign of the response to K is the opposite of the sign in (4). Therefore,

$$\begin{aligned} \theta_K^{\text{net}} \text{ rises with } K &\iff \sigma > \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}, \\ \theta_K^{\text{net}} \text{ is locally unchanged} &\iff \sigma = \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}, \\ \theta_K^{\text{net}} \text{ falls with } K &\iff \sigma < \frac{F_K[1 - \delta(K/Y)]}{F_K - \delta}. \end{aligned} \quad (5)$$

- (b) The net return is $F_K - \delta = 0.05$, so $F_K = 0.08$. Substituting $K/Y = 3$ and $\delta = 0.03$ into (5) gives

$$\sigma > \frac{0.08[1 - (0.03)(3)]}{0.08 - 0.03} = \frac{0.0728}{0.05} = 1.456.$$

Thus

$$\sigma > 1.456$$

is required for a marginal increase in K to raise capital's share of net output.

NOTE SIGN CHECK. The derivative in (4) is taken with respect to F_K , not K . Diminishing marginal returns imply $\partial F_K / \partial K < 0$, so the final inequality reverses.

NOTE GROSS VERSUS NET. F_K is the gross marginal product, whereas $F_K - \delta$ is the net return. Likewise, the relevant denominator is net output $Y - \delta K$, not gross output Y .

PROBLEM 1.12 *Factor-income saving and the golden rule.*

Consider the same setup as at the start of Problem 1.10: the economy is described by the assumptions of the Solow model, except that factors are paid their marginal products and all labor income is consumed and all other income is saved. Show that the economy converges to a balanced growth path, and that the balanced-growth-path level of k equals the golden-rule level of k . What is the intuition for this result?

SOLUTION

Constant returns to scale and marginal-product factor pricing imply that gross capital income is $F_K K$. Since all labor income is consumed and all other income is saved, capital accumulation satisfies

$$\dot{K} = (F_K - \delta)K. \quad (1)$$

Let $k = K/(AL)$. Because AL grows at rate $n + g$ and $F_K = f'(k)$, equation (1) yields

$$\dot{k} = [f'(k) - (n + g + \delta)]k. \quad (2)$$

Apart from the zero-capital steady state, the balanced-growth-path value $k^* > 0$ is determined by

$$f'(k^*) = n + g + \delta. \quad (3)$$

Since $f''(k) < 0$,

$$k < k^* \implies f'(k) > n + g + \delta \implies \dot{k} > 0,$$

$$k > k^* \implies f'(k) < n + g + \delta \implies \dot{k} < 0.$$

Thus every path with positive initial capital converges to k^* . Once k is constant, both K and Y grow at rate $n + g$, so the limiting path is a balanced growth path.

Consumption per unit of effective labor at any constant k is

$$c(k) = f(k) - (n + g + \delta)k.$$

The golden-rule capital stock maximizes $c(k)$, and its first-order condition is

$$f'(k_{GR}) = n + g + \delta. \quad (4)$$

Equations (3) and (4) are identical; hence

$$\boxed{k^* = k_{GR}.$$

The intuition follows directly from saving behavior. Saving per unit of effective labor is capital's gross income, $f'(k)k$, while break-even investment is $(n + g + \delta)k$. Capital rises when the former exceeds the latter and falls when it is smaller. The adjustment ends precisely where the marginal product of capital equals the resource cost of maintaining one more unit of k , which is the golden-rule condition.

NOTE INTERPRETATION. The saving rate is endogenous in this model: households save all capital income and consume all labor income. It should not be treated as the fixed s of the basic Solow model.

PROBLEM 1.13 *The convergence rate of output.*

Go through steps analogous to those in equations (1.29)–(1.32) to find how quickly y converges to y^* in the vicinity of the balanced growth path.

Hint: Since $y = f(k)$, we can write $k = g(y)$, where $g(\cdot) = f^{-1}(\cdot)$.

SOLUTION

Let $k = g(y) = f^{-1}(y)$. Differentiating $y = f(k)$ and using the Solow law of motion gives

$$\begin{aligned}\dot{y} &= f'(k)\dot{k} \\ &= f'(k)[sf(k) - (n + g + \delta)k] \equiv H(y).\end{aligned}\tag{1}$$

At $y^* = f(k^*)$, $H(y^*) = 0$. A first-order approximation around y^* is therefore

$$\dot{y}(t) \simeq H'(y^*)[y(t) - y^*] \equiv -\lambda_y[y(t) - y^*].\tag{2}$$

The solution to the linearized equation is

$$y(t) - y^* \simeq e^{-\lambda_y t}[y(0) - y^*].\tag{3}$$

It remains to calculate $H'(y^*)$. By the chain rule,

$$\begin{aligned}H'(y^*) &= \left. \frac{\partial \dot{y}}{\partial k} \frac{\partial k}{\partial y} \right|_{k=k^*} \\ &= \left\{ f''(k)[sf(k) - (n + g + \delta)k] + f'(k)[sf'(k) - (n + g + \delta)] \right\} \left. \frac{1}{f'(k)} \right|_{k=k^*} \\ &= sf'(k^*) - (n + g + \delta),\end{aligned}\tag{4}$$

where the steady-state condition $sf(k^*) = (n + g + \delta)k^*$ eliminates the first term.

Consequently,

$$\begin{aligned}\lambda_y &= (n + g + \delta) - s f'(k^*) \\ &= (n + g + \delta) \left[1 - \frac{k^* f'(k^*)}{f(k^*)} \right] \\ &= [1 - \alpha_K(k^*)](n + g + \delta).\end{aligned}\tag{5}$$

Thus

$$\lambda_y = \lambda_k = [1 - \alpha_K(k^*)](n + g + \delta).$$

Output per unit of effective labor therefore converges locally at exactly the same rate as capital per unit of effective labor.

NOTE ECONOMIC INTUITION. Near the balanced growth path, $y - y^* \simeq f'(k^*)(k - k^*)$. This local proportionality is why the two gaps shrink at the same exponential rate even though y and k have different units.

PROBLEM 1.14 Embodied technological progress.

(Solow, 1960; Sato, 1966) One view of technological progress is that the productivity of capital goods built at t depends on the state of technology at t and is unaffected by subsequent technological progress. This is known as *embodied technological progress* (technological progress must be “embodied” in new capital before it can raise output). This problem asks you to investigate its effects.

- (a) As a preliminary, let us modify the basic Solow model to make technological progress capital-augmenting rather than labor-augmenting. So that a balanced growth path exists, assume that the production function is Cobb–Douglas:

$$Y(t) = [A(t)K(t)]^\alpha L(t)^{1-\alpha}$$

Assume that A grows at rate μ : $\dot{A}(t) = \mu A(t)$. Show that the economy converges to a balanced growth path, and find the growth rates of Y and K on the balanced growth path.

Hint: Show that we can write $Y/(A^\phi L)$ as a function of $K/(A^\phi L)$, where $\phi = \alpha/(1 - \alpha)$. Then analyze the dynamics of $K/(A^\phi L)$.

- (b) Now consider embodied technological progress. Specifically, let the production function be $Y(t) = J(t)^\alpha L(t)^{1-\alpha}$, where $J(t)$ is the effective capital stock. The dynamics of $J(t)$ are given by $\dot{J}(t) = s A(t) Y(t) - \delta J(t)$. The presence of the $A(t)$ term in this expression means that the productivity of investment at t depends on the technology at t . Show that the economy converges to a balanced growth path. What are the growth rates of Y and J on the balanced growth path?

Hint: Let $\bar{J}(t) \equiv J(t)/A(t)$. Then use the same approach as in (a), focusing on $\bar{J}/(A^\phi L)$ instead of $K/(A^\phi L)$.

- (c) What is the elasticity of output on the balanced growth path with respect to s ?
- (d) In the vicinity of the balanced growth path, how rapidly does the economy converge to the balanced growth path?
- (e) Compare your results for (c) and (d) with the corresponding results in the text for the basic Solow model.

SOLUTION

Let

$$\phi \equiv \frac{\alpha}{1-\alpha}.$$

- (a) Define capital and output relative to $A^\phi L$:

$$\kappa(t) \equiv \frac{K(t)}{A(t)^\phi L(t)}, \quad y(t) \equiv \frac{Y(t)}{A(t)^\phi L(t)}.$$

Since $1 + \phi = 1/(1 - \alpha)$, dividing the production function by $A^\phi L$ gives

$$y(t) = \kappa(t)^\alpha. \quad (1)$$

The normalizing factor $A^\phi L$ grows at rate $\phi\mu + n$. Using $\dot{K} = sY - \delta K$, the law of motion for κ is

$$\dot{\kappa} = s\kappa^\alpha - (n + \phi\mu + \delta)\kappa. \quad (2)$$

The unique positive steady state is

$$\kappa^* = \left(\frac{s}{n + \phi\mu + \delta} \right)^{1/(1-\alpha)}. \quad (3)$$

For $0 < \kappa < \kappa^*$, actual investment exceeds break-even investment and $\dot{\kappa} > 0$; for $\kappa > \kappa^*$, the reverse holds. Thus $\kappa \rightarrow \kappa^*$ and, by (1), $y \rightarrow y^*$.

Since $K = A^\phi L \kappa$ and $Y = A^\phi L y$, the balanced-growth-path growth rates are

$$\boxed{g_K^{\text{BGP}} = g_Y^{\text{BGP}} = n + \phi\mu.}$$

- (b) Under embodied technological progress, define $\bar{J} \equiv J/A$. Its law of motion is

$$\dot{\bar{J}} = \frac{\dot{J}}{A} - \mu\bar{J} = sY - (\delta + \mu)\bar{J}. \quad (4)$$

The production function becomes $Y = (A\bar{J})^\alpha L^{1-\alpha}$, which has the same form as in

part (a). Define

$$j(t) \equiv \frac{\bar{J}(t)}{A(t)^\phi L(t)} = \frac{J(t)}{A(t)^{1+\phi} L(t)}.$$

Then

$$y(t) = j(t)^\alpha, \quad y(t) \equiv \frac{Y(t)}{A(t)^\phi L(t)}. \quad (5)$$

Combining (4) with the growth rate of $A^\phi L$ yields

$$\dot{j} = sj^\alpha - [n + \delta + (1 + \phi)\mu]j. \quad (6)$$

Thus the unique positive steady state is

$$j^* = \left(\frac{s}{n + \delta + (1 + \phi)\mu} \right)^{1/(1-\alpha)}, \quad (7)$$

and the same sign argument as in part (a) shows that $j \rightarrow j^*$.

On the balanced growth path, y and j are constant. Hence

$$\boxed{g_Y^{\text{BGP}} = n + \phi\mu,} \quad \boxed{g_J^{\text{BGP}} = n + (1 + \phi)\mu.}$$

The growth rate of J is higher because $J = A^{1+\phi} L j$, whereas output is $Y = A^\phi L y$.

(c) From (5) and (7),

$$y^* = \left(\frac{s}{n + \delta + (1 + \phi)\mu} \right)^{\alpha/(1-\alpha)}.$$

Since $Y = A^\phi L y$, the elasticity of balanced-growth-path output with respect to s is

$$\boxed{\frac{\partial \ln Y^{\text{BGP}}}{\partial \ln s} = \frac{\partial \ln y^*}{\partial \ln s} = \frac{\alpha}{1 - \alpha}.} \quad (8)$$

(d) Let $B \equiv n + \delta + (1 + \phi)\mu$. Linearizing (6) around j^* gives

$$\dot{j} \simeq [s\alpha(j^*)^{\alpha-1} - B](j - j^*).$$

The steady-state condition implies $s(j^*)^{\alpha-1} = B$. Therefore,

$$\boxed{\lambda_{\text{emb}} = (1 - \alpha)[n + \delta + (1 + \phi)\mu].} \quad (9)$$

Locally,

$$j(t) - j^* \simeq e^{-\lambda_{\text{emb}} t} [j(0) - j^*].$$

Because $y = j^\alpha$ is smooth and strictly increasing, y converges at the same rate.

(e) For Cobb–Douglas production, the basic Solow model gives

$$\frac{\partial \ln y^*}{\partial \ln s} = \frac{\alpha}{1 - \alpha}, \quad \lambda_{\text{basic}} = (1 - \alpha)(n + \mu + \delta).$$

Thus embodiment leaves the saving elasticity unchanged. It increases the convergence rate, since

$$\lambda_{\text{emb}} - \lambda_{\text{basic}} = \alpha\mu > 0.$$

The two convergence mechanisms are summarized in the figure below. Both use the same concave actual-investment schedule; embodiment adds the extra $\mu\bar{j}$ dilution term visible in (4), making the break-even line steeper in normalized units.

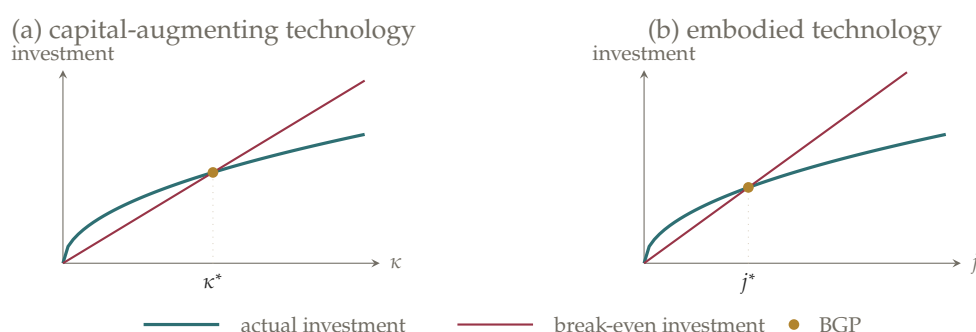


Figure 1.8. Normalized dynamics under capital-augmenting and embodied technological progress. The break-even schedules are $(n + \phi\mu + \delta)\kappa$ and $[n + \delta + (1 + \phi)\mu]j$, respectively.

NOTE NOTATION. J is the effective capital stock, $\bar{J} = J/A$ removes the productivity of the investment vintage, and $j = J/(A^{1+\phi}L)$ is the stationary normalized variable. Mixing these three objects leads to incorrect growth rates.

NOTE ECONOMIC INTUITION. As the frontier A rises, existing embodied capital loses ground relative to newly produced capital goods. This adds μ to normalized break-even investment and makes convergence faster than in the basic model.

PROBLEM 1.15 *Growth accounting versus growth causality.*

Consider a Solow economy on its balanced growth path. Suppose the growth-accounting techniques described in Section 1.7 are applied to this economy.

- What fraction of growth in output per worker does growth accounting attribute to growth in capital per worker? What fraction does it attribute to technological progress?
- How can you reconcile your results in (a) with the fact that the Solow model implies that the growth rate of output per worker on the balanced growth path is determined solely by the rate of technological progress?

SOLUTION

Let $\alpha_K(t) \equiv F_K(t)K(t)/Y(t)$ denote capital's output elasticity. Growth accounting for output per worker gives

$$\frac{\dot{Y}}{Y} - \frac{\dot{L}}{L} = \alpha_K(t) \left(\frac{\dot{K}}{K} - \frac{\dot{L}}{L} \right) + R(t), \quad (1)$$

where $R(t)$ is the Solow residual.

(a) On the Solow balanced growth path,

$$\frac{\dot{Y}}{Y} - \frac{\dot{L}}{L} = \frac{\dot{K}}{K} - \frac{\dot{L}}{L} = g. \quad (2)$$

Substituting (2) into (1) gives

$$g = \alpha_K(k^*)g + R, \quad R = [1 - \alpha_K(k^*)]g. \quad (3)$$

Thus growth accounting attributes

$$\alpha_K(k^*) \text{ of growth to capital deepening}$$

and

$$1 - \alpha_K(k^*) \text{ of growth to technological progress.}$$

With the common calibration $\alpha_K = 1/3$, the shares are one-third and two-thirds, respectively.

(b) There is no contradiction because growth accounting records *proximate* contributions, whereas the Solow model identifies the *underlying* source of sustained growth. Growth in A raises output per worker directly. For a fixed saving rate, it also induces capital per worker to grow at rate g , so part of technology's total effect operates through capital deepening. Growth accounting assigns this induced channel to capital rather than to the technological change that caused it.

Hence the accounting decomposition is correct, but it should not be read as a causal decomposition of the ultimate sources of balanced-growth-path growth.

NOTE ACCOUNTING VERSUS CAUSALITY. The term $\alpha_K g$ answers how much current capital deepening contributes mechanically to output growth. It does not answer why capital deepening occurs.

NOTE SOLOW RESIDUAL. Labor-augmenting technology grows at rate g , but the residual in (3) is $(1 - \alpha_K)g$. The remaining $\alpha_K g$ appears in the capital-deepening term.

PROBLEM 1.16 *Measurement error in convergence regressions.*

- (a) In the model of convergence and measurement error in equations (1.39) and (1.40), suppose the true value of b is -1 . Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b ? Explain.
- (b) Suppose there is measurement error in measured 1979 income per capita but not in 1870 income per capita. Does a regression of $\ln(Y/N)_{1979} - \ln(Y/N)_{1870}$ on a constant and $\ln(Y/N)_{1870}$ yield a biased estimate of b ? Explain.

SOLUTION

Write true log income per capita in 1870 as x_i^* , measured log income as x_i , and true 1979 log income as z_i^* .

- (a) Equations (1.39) and (1.40) are

$$z_i^* - x_i^* = a + bx_i^* + \varepsilon_i, \quad (1)$$

$$x_i = x_i^* + u_i, \quad (2)$$

where u_i is uncorrelated with x_i^* and ε_i . The regression uses measured growth $z_i^* - x_i$. Substituting $x_i^* = x_i - u_i$ into (1) gives

$$z_i^* - x_i = a + bx_i + \varepsilon_i - (1 + b)u_i. \quad (3)$$

In general, $x_i = x_i^* + u_i$ is correlated with the composite error. Indeed,

$$\text{Cov}(x_i, \varepsilon_i - (1 + b)u_i) = -(1 + b)\text{Var}(u_i). \quad (4)$$

When the true $b = -1$, however, the measurement-error term disappears from (3); the regression error is simply ε_i , which is uncorrelated with x_i . Therefore,

$b = -1 \implies \text{the OLS slope is unbiased.}$

- (b) Now suppose 1870 income is measured without error, so $x_i = x_i^*$, but measured 1979 income is $z_i = z_i^* + v_i$. Measured growth satisfies

$$z_i - x_i = a + bx_i + \varepsilon_i + v_i. \quad (5)$$

Under classical measurement error, $E[v_i | x_i] = 0$. Thus the new composite error $\varepsilon_i + v_i$ remains uncorrelated with the regressor, and

measurement error only in 1979 income does not bias $\hat{b}$.

It raises the variance of the regression error and therefore reduces precision.

NOTE WHY INITIAL-INCOME ERROR IS SPECIAL. Measured 1870 income enters both sides with opposite signs. A positive error makes a country look richer initially and slower growing, mechanically favoring convergence unless $b = -1$.

NOTE CLASSICAL-ERROR QUALIFICATION. Part (b) requires mean-zero 1979 error uncorrelated with initial income and the structural disturbance. Systematic error can still bias OLS.

PROBLEM 1.17 *Growth without resource and land limitations.*

Derive equation (1.50).

Hint: Follow steps analogous to those in equations [1.47] and [1.48].

SOLUTION

The hypothetical economy removes resource and land limitations by assuming that resources, land, and labor all grow at rate n :

$$g_R = g_T = g_L = n, \quad g_A = g.$$

The capital accumulation equation gives $g_K = s(Y/K) - \delta$. A constant balanced-growth-path value of g_K therefore requires Y/K to be constant, and hence

$$g_K^{\text{BGP}} = g_Y^{\text{BGP}}. \quad (1)$$

Log-differentiating the production function gives

$$g_Y = \alpha g_K + \beta g_R + \gamma g_T + (1 - \alpha - \beta - \gamma)(g_A + g_L). \quad (2)$$

Substitute $g_R = g_T = g_L = n$ and $g_A = g$:

$$\begin{aligned} g_Y &= \alpha g_K + (\beta + \gamma)n + (1 - \alpha - \beta - \gamma)(n + g) \\ &= \alpha g_K + (1 - \alpha)n + (1 - \alpha - \beta - \gamma)g. \end{aligned} \quad (3)$$

Imposing (1) yields

$$\begin{aligned} (1 - \alpha)\tilde{g}_Y^{\text{BGP}} &= (1 - \alpha)n + (1 - \alpha - \beta - \gamma)g, \\ \tilde{g}_Y^{\text{BGP}} &= n + \frac{1 - \alpha - \beta - \gamma}{1 - \alpha}g. \end{aligned} \quad (4)$$

The tilde distinguishes the hypothetical economy from the economy with fixed land and declining resource use. Output per worker grows at total-output growth minus population growth. Therefore,

$$\boxed{\tilde{g}_{Y/L}^{\text{BGP}} = \tilde{g}_Y^{\text{BGP}} - n = \frac{1 - \alpha - \beta - \gamma}{1 - \alpha}g.} \quad (5)$$

This is equation (1.50).

CHAPTER 2

PROBLEM 2.1

Consider N firms each with the constant-returns-to-scale production function $Y = F(K, AL)$, or (using the intensive form) $Y = AL f(k)$. Assume $f'(\cdot) > 0$, $f''(\cdot) < 0$. Assume that all firms can hire labor at wage wA and rent capital at cost r , and that all firms have the same value of A .

- Consider the problem of a firm trying to produce Y units of output at minimum cost. Show that the cost-minimizing level of k is uniquely defined and is independent of Y , and that all firms therefore choose the same value of k .
- Show that the total output of the N cost-minimizing firms equals the output that a single firm with the same production function has if it uses all the labor and capital used by the N firms.

PROBLEM 2.2 *The elasticity of substitution with constant-relative-risk-aversion utility.*

Consider an individual who lives for two periods and whose utility is given by equation (2.43). Let P_1 and P_2 denote the prices of consumption in the two periods, and let W denote the value of the individual's lifetime income; thus the budget constraint is $P_1 C_1 + P_2 C_2 = W$.

- What are the individual's utility-maximizing choices of C_1 and C_2 , given P_1 , P_2 , and W ?
- The elasticity of substitution between consumption in the two periods is

$$-\frac{P_1/P_2}{C_1/C_2} \frac{\partial (C_1/C_2)}{\partial (P_1/P_2)} \quad \text{or} \quad -\frac{\partial \ln (C_1/C_2)}{\partial \ln (P_1/P_2)}$$

Show that with the utility function (2.43), the elasticity of substitution between C_1 and C_2 is $1/\theta$.

PROBLEM 2.3

- Suppose it is known in advance that at some time t_0 the government will confiscate half of whatever wealth each household holds at that time. Does consumption

change discontinuously at time t_0 ? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

- (b) Suppose it is known in advance that at t_0 the government will confiscate from each household an amount of wealth equal to half of the wealth of the average household at that time. Does consumption change discontinuously at time t_0 ? If so, why (and what is the condition relating consumption immediately before t_0 to consumption immediately after)? If not, why not?

PROBLEM 2.4

Assume that the instantaneous utility function $u(C)$ in equation (2.2) is $\ln C$. Consider the problem of a household maximizing (2.2) subject to (2.7). Find an expression for C at each time as a function of initial wealth plus the present value of labor income, the path of $r(t)$, and the parameters of the utility function.

PROBLEM 2.5

Consider a household with utility given by (2.2)–(2.3). Assume that the real interest rate is constant, and let W denote the household's initial wealth plus the present value of its lifetime labor income (the right-hand side of [2.7]). Find the utility-maximizing path of C , given r , W , and the parameters of the utility function.

PROBLEM 2.6 *Growth, saving, and $r - g$.*

Piketty (2014) argues that a fall in the growth rate of the economy is likely to lead to an increase in the difference between the real interest rate and the growth rate. This problem asks you to investigate this issue in the context of the Ramsey–Cass–Koopmans model. Specifically, consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path, and suppose there is a permanent fall in g .

- How, if at all, does this affect the $\dot{k} = 0$ curve?
- How, if at all, does this affect the $\dot{c} = 0$ curve?
- At the time of the change, does c rise, fall, or stay the same, or is it not possible to tell?
- At the time of the change, does $r - g$ rise, fall, or stay the same, or is it not possible to tell?
- In the long run, does $r - g$ rise, fall, or stay the same, or is it not possible to tell?
- Find an expression for the impact of a marginal change in g on the fraction of out-

put that is saved on the balanced growth path. Can one tell whether this expression is positive or negative?

- (g) For the case where the production function is Cobb–Douglas, $f(k) = k^\alpha$, rewrite your answer to part (f) in terms of ρ , n , g , θ , and α .

Hint: Use the fact that $f'(k^*) = \rho + \theta g$.

PROBLEM 2.7

Describe how each of the following affects the $\dot{c} = 0$ and $\dot{k} = 0$ curves in Figure 2.5, and thus how they affect the balanced-growth-path values of c and k :

- (a) A rise in θ .
- (b) A downward shift of the production function.
- (c) A change in the rate of depreciation from the value of zero assumed in the text to some positive level.

PROBLEM 2.8

Derive an expression analogous to (2.40) for the case of a positive depreciation rate.

PROBLEM 2.9 *A closed-form solution of the Ramsey model.*

(Smith, 2006) Consider the Ramsey model with Cobb–Douglas production, $y(t) = k(t)^\alpha$, and with the coefficient of relative risk aversion (θ) and capital's share (α) assumed to be equal.

- (a) What is k on the balanced growth path (k^*)?
- (b) What is c on the balanced growth path (c^*)?
- (c) Let $z(t)$ denote the capital-output ratio,

$$k(t)/y(t)$$

and $x(t)$ denote the consumption-capital ratio,

$$c(t)/k(t)$$

Find expressions for $\dot{z}(t)$ and $\dot{x}(t)/x(t)$ in terms of z , x , and the parameters of the model.

- (d) Tentatively conjecture that x is constant along the saddle path. Given this conjecture:

- (i) Find the path of z given its initial value, $z(0)$.
- (ii) Find the path of y given the initial value of k , $k(0)$. Is the speed of convergence to the balanced growth path, $d \ln [y(t) - y^*] / dt$, constant as the economy moves along the saddle path?
- (e) In the conjectured solution, are the equations of motion for c and k , (2.25) and (2.26), satisfied?

PROBLEM 2.10 *Capital taxation in the Ramsey–Cass–Koopmans model.*

Consider a Ramsey–Cass–Koopmans economy that is on its balanced growth path. Suppose that at some time, which we will call time 0, the government switches to a policy of taxing investment income at rate τ . Thus the real interest rate that households face is now given by $r(t) = (1 - \tau) f'(k(t))$. Assume that the government returns the revenue it collects from this tax through lump-sum transfers. Finally, assume that this change in tax policy is unanticipated.

- (a) How, if at all, does the tax affect the $\dot{c} = 0$ locus? The $\dot{k} = 0$ locus?
- (b) How does the economy respond to the adoption of the tax at time 0? What are the dynamics after time 0?
- (c) How do the values of c and k on the new balanced growth path compare with their values on the old balanced growth path?
- (d) (Barro, Mankiw, and Sala-i-Martin, 1995) Suppose there are many economies like this one. Workers' preferences are the same in each country, but the tax rates on investment income may vary across countries. Assume that each country is on its balanced growth path.
 - (i) Show that the saving rate on the balanced growth path, $(y^* - c^*) / y^*$, is decreasing in τ .
 - (ii) Do citizens in low- τ , high- k^* , high-saving countries have any incentive to invest in low-saving countries? Why or why not?
- (e) Does your answer to part (c) imply that a policy of *subsidizing investment* (that is, making $\tau < 0$), and raising the revenue for this subsidy through lump-sum taxes, increases welfare? Why or why not?
- (f) How, if at all, do the answers to parts (a) and (b) change if the government does not rebate the revenue from the tax but instead uses it to make government purchases?

PROBLEM 2.11 *Using the phase diagram to analyze the impact of an anticipated change.*

Consider the policy described in Problem 2.10, but suppose that instead of announcing and implementing the tax at time 0, the government announces at time 0 that at some later time, time t_1 , investment income will begin to be taxed at rate τ .

- (a) Draw the phase diagram showing the dynamics of c and k after time t_1 .
- (b) Can c change discontinuously at time t_1 ? Why or why not?
- (c) Draw the phase diagram showing the dynamics of c and k before t_1 .
- (d) In light of your answers to parts (a), (b), and (c), what must c do at time 0?
- (e) Summarize your results by sketching the paths of c and k as functions of time.

PROBLEM 2.12 *Using the phase diagram to analyze the impact of unanticipated and anticipated temporary changes.*

Analyze the following two variations on Problem 2.11:

- (a) At time 0, the government announces that it will tax investment income at rate τ from time 0 until some later date t_1 ; thereafter investment income will again be untaxed.
- (b) At time 0, the government announces that from time t_1 to some later time t_2 , it will tax investment income at rate τ ; before t_1 and after t_2 , investment income will not be taxed.

PROBLEM 2.13 *An interesting situation in the Ramsey–Cass–Koopmans model.*

- (a) Consider the Ramsey–Cass–Koopmans model where k at time 0 (which—as always—the model takes as given) is at the golden-rule level: $k(0) = k^{GR}$. Sketch the paths of c and k .
- (b) Consider the same initial situation as in part (a), but in the version of the model that includes government purchases. Assume that G is constant and equal $\bar{G}$. Crucially, $\bar{G}$ is strictly less than $f(k^{GR}) - (n + g)k^{GR}$ and strictly greater than $f(k^*) - (n + g)k^*$ (where k^* is the level of k on the balanced growth path the economy would have if G were constant and equal to 0). Sketch the paths of c and k .

Hint: This problem is hard but can be answered by thinking things through slowly and carefully.

PROBLEM 2.14

Consider the Diamond model with logarithmic utility and Cobb–Douglas production. Describe how each of the following affects k_{t+1} as a function of k_t :

- (a) A rise in n .
- (b) A downward shift of the production function (that is, $f(k)$ takes the form Bk^α , and B falls).
- (c) A rise in α .

PROBLEM 2.15 *A discrete-time version of the Solow model.*

Suppose $Y_t = F(K_t, A_t L_t)$, with $F(\cdot)$ having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1 + g) A_t$, $L_{t+1} = (1 + n) L_t$, and $K_{t+1} = K_t + s Y_t - \delta K_t$.

- (a) Find an expression for k_{t+1} as a function of k_t .
- (b) Sketch k_{t+1} as a function of k_t . Does the economy have a balanced growth path? If the initial level of k differs from the value on the balanced growth path, does the economy converge to the balanced growth path?
- (c) Find an expression for consumption per unit of effective labor on the balanced growth path as a function of the balanced-growth-path value of k . What is the marginal product of capital, $f'(k)$, when k maximizes consumption per unit of effective labor on the balanced growth path?
- (d) Assume that the production function is Cobb–Douglas.
 - (i) What is k_{t+1} as a function of k_t ?
 - (ii) What is k^* , the value of k on the balanced growth path?
 - (iii) Along the lines of equations (2.65)–(2.67), in the text, linearize the expression in subpart (i) around $k_t = k^*$, and find the rate of convergence of k to k^* .

PROBLEM 2.16 *Depreciation in the Diamond model and microeconomic foundations for the Solow model.*

Suppose that in the Diamond model capital depreciates at rate δ , so that $r_t = f'(k_t) - \delta$.

- (a) How, if at all, does this change in the model affect equation (2.60) giving k_{t+1} as a function of k_t ?
- (b) In the special case of logarithmic utility, Cobb–Douglas production, and $\delta = 1$, what is the equation for k_{t+1} as a function of k_t ? Compare this with the analogous

expression for the discrete-time version of the Solow model with $\delta = 1$ from part (a) of Problem 2.15.

PROBLEM 2.17 *Social security in the Diamond model.*

Consider a Diamond economy where g is zero, production is Cobb–Douglas, and utility is logarithmic.

- (a) **Pay-as-you-go social security.** Suppose the government taxes each young individual an amount T and uses the proceeds to pay benefits to old individuals; thus each old person receives $(1 + n)T$.
- (i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t ?
 - (ii) How, if at all, does this change affect the balanced-growth-path value of k ?
 - (iii) If the economy is initially on a balanced growth path that is dynamically efficient, how does a marginal increase in T affect the welfare of current and future generations? What happens if the initial balanced growth path is dynamically inefficient?
- (b) **Fully funded social security.** Suppose the government taxes each young person an amount T and uses the proceeds to purchase capital. Individuals born at t therefore receive $(1 + r_{t+1})T$ when they are old.
- (i) How, if at all, does this change affect equation (2.61) giving k_{t+1} as a function of k_t ?
 - (ii) How, if at all, does this change affect the balanced-growth-path value of k ?

PROBLEM 2.18 *The basic overlapping-generations model.*

(Samuelson, 1958; Allais, 1947) Suppose, as in the Diamond model, that L_t two-period-lived individuals are born in period t and that $L_t = (1 + n)L_{t-1}$. For simplicity, let utility be logarithmic with no discounting: $U_t = \ln(C_{1t}) + \ln(C_{2t+1})$.

The production side of the economy is simpler than in the Diamond model. Each individual born at time t is endowed with A units of the economy's single good. The good can be either consumed or stored. Each unit stored yields $x > 0$ units of the good in the following period.^[1]

Finally, assume that in the initial period, period 0, in addition to the L_0 young indi-

[1] Note that this is the same as the Diamond economy with $g = 0$, $F(K_t, AL_t) = AL_t + xK_t$, and $\delta = 1$. With this production function, since individuals supply 1 unit of labor when they are young, an individual born in t obtains A units of the good. And each unit saved yields $1 + r = 1 + \partial F(K, AL) / \partial K - \delta = 1 + x - 1 = x$ units of second-period consumption.

viduals each endowed with A units of the good, there are $[1/(1+n)]L_0$ individuals who are alive only in period 0. Each of these “old” individuals is endowed with some amount Z of the good; their utility is simply their consumption in the initial period, C_{20} .

- (a) Describe the decentralized equilibrium of this economy.

Hint: Given the overlapping-generations structure, will the members of any generation engage in transactions with members of another generation?

- (b) Consider paths where the fraction of agents’ endowments that is stored, f_t , is constant over time. What is total consumption (that is, consumption of all the young plus consumption of all the old) per person on such a path as a function of f ? If $x < 1+n$, what value of f satisfying $0 \leq f \leq 1$ maximizes consumption per person? Is the decentralized equilibrium Pareto efficient in this case? If not, how can a social planner raise welfare?

PROBLEM 2.19 *Stationary monetary equilibria in the Samuelson OLG model.*

(Samuelson, 1958) Consider the setup described in Problem 2.18. Assume that $x < 1+n$. Suppose that the old individuals in period 0, in addition to being endowed with Z units of the good, are each endowed with M units of a storable, divisible commodity, which we will call money. Money is not a source of utility.

- (a) Consider an individual born at t . Suppose the price of the good in units of money is P_t in t and P_{t+1} in $t+1$. Thus the individual can sell units of endowment for P_t units of money and then use that money to buy P_t/P_{t+1} units of the next generation’s endowment the following period. What is the individual’s behavior as a function of P_t/P_{t+1} ?
- (b) Show that there is an equilibrium with $P_{t+1} = P_t/(1+n)$ for all $t \geq 0$ and no storage, and thus that the presence of “money” allows the economy to reach the golden-rule level of storage.
- (c) Show that there are also equilibria with $P_{t+1} = P_t/x$ for all $t \geq 0$.
- (d) Finally, explain why $P_t = \infty$ for all t (that is, money is worthless) is also an equilibrium. Explain why this is the only equilibrium if the economy ends at some date, as in Problem 2.20(b) below.

Hint: Reason backward from the last period.

PROBLEM 2.20 *The source of dynamic inefficiency.*

(Shell, 1971) There are two ways in which the Diamond and Samuelson models differ from textbook models. First, markets are incomplete: because individuals cannot trade with individuals who have not been born, some possible transactions are ruled out. Second, because time goes on forever, there are an infinite number of agents. This problem asks you to investigate which of these is the source of the possibility of dynamic inefficiency. For simplicity, it focuses on the Samuelson overlapping-generations model (see the previous two problems), again with log utility and no discounting. To simplify further, it assumes $n = 0$ and $0 < x < 1$.

- (a) **Incomplete markets.** Suppose we eliminate incomplete markets from the model by allowing all agents to trade in a competitive market “before” the beginning of time. That is, a Walrasian auctioneer calls out prices $Q_0, Q_1, Q_2, \dots$ for the good at each date. Individuals can then make sales and purchases at these prices given their endowments and their ability to store. The budget constraint of an individual born at t is thus $Q_t C_{1t} + Q_{t+1} C_{2t+1} = Q_t (A - S_t) + Q_{t+1} x S_t$, where S_t (which must satisfy $0 \leq S_t \leq A$) is the amount the individual stores.
- (i) Suppose the auctioneer announces $Q_{t+1} = Q_t/x$ for all $t > 0$. Show that in this case individuals are indifferent concerning how much to store, that there is a set of storage decisions such that markets clear at every date, and that this equilibrium is the same as the equilibrium described in part (a) of Problem 2.18.
- (ii) Suppose the auctioneer announces prices that fail to satisfy $Q_{t+1} = Q_t/x$ at some date. Show that at the first date that does not satisfy this condition the market for the good cannot clear, and thus that the proposed price path cannot be an equilibrium.
- (b) **Infinite duration.** Suppose that the economy ends at some date T . That is, suppose the individuals born at T live only one period (and hence seek to maximize C_{1T}), and that thereafter no individuals are born. Show that the decentralized equilibrium is Pareto efficient.
- (c) In light of these answers, is it incomplete markets or infinite duration that is the source of dynamic inefficiency?

PROBLEM 2.21 *Explosive paths in the Samuelson overlapping-generations model.*

(Black, 1974; Brock, 1975; Calvo, 1978) Consider the setup described in Problem 2.19. Assume that x is zero, and assume that utility is constant-relative-risk-aversion with $\theta < 1$ rather than logarithmic. Finally, assume for simplicity that $n = 0$.

- (a) What is the behavior of an individual born at t as a function of P_t/P_{t+1} ? Show

that the amount of his or her endowment that the individual sells for money is an increasing function of P_t/P_{t+1} and approaches zero as this ratio approaches zero.

- (b) Suppose $P_0/P_1 < 1$. How much of the good are the individuals born in period 0 planning to buy in period 1 from the individuals born then? What must P_1/P_2 be for the individuals born in period 1 to want to supply this amount?
- (c) Iterating this reasoning forward, what is the qualitative behavior of P_t/P_{t+1} over time? Does this represent an equilibrium path for the economy?
- (d) Can there be an equilibrium path with $P_0/P_1 > 1$?

CHAPTER 3

PROBLEM 3.1

Consider the model of Section 3.2 with $\theta < 1$.

- On the balanced growth path, $\dot{A} = g_A^* A(t)$, where g_A^* is the balanced-growth-path value of g_A . Use this fact and equation (3.6) to derive an expression for $A(t)$ on the balanced growth path in terms of B , a_L , γ , θ , and $L(t)$.
- Use your answer to part (a) and the production function, (3.5), to obtain an expression for $Y(t)$ on the balanced growth path. Find the value of a_L that maximizes output on the balanced growth path.

PROBLEM 3.2

Consider two economies (indexed by $i = 1, 2$) described by $Y_i(t) = K_i(t)^\theta$ and $\dot{K}_i(t) = s_i Y_i(t)$, where $\theta > 1$. Suppose that the two economies have the same initial value of K , but that $s_1 > s_2$. Show that Y_1/Y_2 is continually rising.

PROBLEM 3.3

Consider the economy analyzed in Section 3.3. Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:

- An increase in n .
- An increase in a_K .
- An increase in θ .

PROBLEM 3.4

Consider the economy described in Section 3.3, and assume $\beta + \theta < 1$ and $n > 0$. Suppose the economy is initially on its balanced growth path, and that there is a permanent increase in s .

- How, if at all, does the change affect the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines? How, if at

all, does it affect the location of the economy in (g_A, g_K) space at the time of the change?

- (b) What are the dynamics of g_A and g_K after the increase in s ? Sketch the path of log output per worker.
- (c) Intuitively, how does the effect of the increase in s compare with its effect in the Solow model?

PROBLEM 3.5

Consider the model of Section 3.3 with $\beta + \theta = 1$ and $n = 0$.

- (a) Using (3.14) and (3.16), find the value that A/K must have for g_K and g_A to be equal.
- (b) Using your result in part (a), find the growth rate of A and K when $g_K = g_A$.
- (c) How does an increase in s affect the long-run growth rate of the economy?
- (d) What value of a_K maximizes the long-run growth rate of the economy? Intuitively, why is this value not increasing in β , the importance of capital in the R&D sector?

PROBLEM 3.6

Consider the model of Section 3.3 with $\beta + \theta > 1$ and $n > 0$.

- (a) Draw the phase diagram for this case.
- (b) Show that regardless of the economy's initial conditions, eventually the growth rates of A and K (and hence the growth rate of Y) are increasing continually.
- (c) Repeat parts (a) and (b) for the case of $\beta + \theta = 1, n > 0$.

PROBLEM 3.7 *Learning-by-doing.*

Suppose that output is given by equation (3.22), $Y(t) = K(t)^\alpha [A(t)L(t)]^{1-\alpha}$; that L is constant and equal to 1; that $\dot{K}(t) = sY(t)$; and that knowledge accumulation occurs as a side effect of goods production: $\dot{A}(t) = BY(t)$.

- (a) Find expressions for $g_A(t)$ and $g_K(t)$ in terms of $A(t)$, $K(t)$, and the parameters.
- (b) Sketch the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines in (g_A, g_K) space.
- (c) Does the economy converge to a balanced growth path? If so, what are the growth rates of K , A , and Y on the balanced growth path?
- (d) How does an increase in s affect long-run growth?

PROBLEM 3.8

Consider the model of Section 3.5. Suppose, however, that households have constant-relative-risk-aversion utility with a coefficient of relative risk aversion of θ . Find the equilibrium level of labor in the R&D sector, L_A .

PROBLEM 3.9

Suppose that policymakers, realizing that monopoly power creates distortions, put controls on the prices that patent-holders in the Romer model can charge for the inputs embodying their ideas. Specifically, suppose they require patent-holders to charge $\delta w(t)/\phi$, where δ satisfies $\phi \leq \delta \leq 1$.

- (a) What is the equilibrium growth rate of the economy as a function of δ and the other parameters of the model? Does a reduction in δ increase, decrease, or have no effect on the equilibrium growth rate, or is it not possible to tell?
- (b) Explain intuitively why setting $\delta = \phi$, thereby requiring patent-holders to charge marginal cost and so eliminating the monopoly distortion, does not maximize social welfare.

PROBLEM 3.10

- (a) Show that (3.48) follows from (3.47).
- (b) Derive (3.49).

PROBLEM 3.11 *The balanced growth path of a semi-endogenous version of the Romer model.*

(Jones, 1995) Consider the model of Section 3.5 with two changes. First, existing knowledge contributes less than proportionally to the production of new knowledge, as in Case 1 of the model of Section 3.2: $\dot{A}(t) = BL_A(t)A(t)^\theta$, $\theta < 1$. Second, population is growing at rate n rather than constant: $L(t) = L(0)e^{nt}$, $n > 0$. (Consistent with this, assume that utility is given by equation [2.2] with $u(\cdot)$ logarithmic.) The analysis of Section 3.2 implies that such a model will exhibit transition dynamics rather than always immediately being on a balanced growth path. This problem therefore asks you not to analyze the full dynamics of the model, but to focus on its properties when it is on a balanced growth path. Specifically, it looks at situations where the fraction of the labor force engaged in R&D, a_L , is constant, and all variables of the model are growing at constant rates.

Hint: You are welcome to assume rather than derive that on a balanced growth path, the wage and consumption per person grow at the same rate as Y/L .

- (a) What are balanced-growth-path values of the growth rates of Y and Y/L and of

r as functions of the balanced-growth-path value of the growth rate of A and parameters of the model?

- (b) On the balanced growth path, what is the present value of the profits from the discovery of a new idea at time t as a function of $L(t)$, $A(t)$, $w(t)$, and exogenous parameters?
- (c) What is $\dot{A}(t)/A(t)$ on the balanced growth path as a function of a_L and exogenous parameters? What is $L(t)A(t)^{\theta-1}$ on the balanced growth path?

Hint: Consider equations [3.6] and [3.7] in the case of $\gamma = 1$.

- (d) What is a_L on the balanced growth path?
- (e) Discuss how changes in each of ρ , B , ϕ , n , and θ affect the balanced-growth-path value of a_L . In the cases of ρ , B , and ϕ , are the effects in the same direction as the effects on L_A in the model of Section 3.5?

PROBLEM 3.12 *Learning-by-doing with microeconomic foundations.*

Consider a variant of the model in equations (3.22)–(3.25). Suppose firm i 's output is $Y_i(t) = K_i(t)^\alpha [A(t)L_i(t)]^{1-\alpha}$, and that $A(t) = BK(t)$. Here K_i and L_i are the amounts of capital and labor used by firm i , and K is the aggregate capital stock. Capital and labor earn their *private* marginal products. As in the model of Section 3.5, the economy is populated by infinitely lived households that own the economy's initial capital stock. The utility of the representative household takes the constant-relative-risk-aversion form in equations (2.2)–(2.3). Population growth is zero.

- (a) (i) What are the private marginal products of capital and labor at firm i as functions of $K_i(t)$, $L_i(t)$, $K(t)$, and the parameters of the model?
- (ii) Explain why the capital-labor ratio must be the same at all firms, so

$$\frac{K_i(t)}{L_i(t)} = \frac{K(t)}{L(t)}$$

- (iii) What are $w(t)$ and $r(t)$ as functions of $K(t)$, L , and the parameters of the model?
- (b) What must the growth rate of consumption be in equilibrium? Assume for simplicity that the parameter values are such that the growth rate is strictly positive and less than the interest rate. Sketch an explanation of why the equilibrium growth rate of output equals the equilibrium growth rate of consumption.

Hint: Consider equation [2.22].

- (c) Describe how long-run growth is affected by:

- (i) A rise in B .
- (ii) A rise in ρ .
- (iii) A rise in L .
- (d) Is the equilibrium growth rate more than, less than, or equal to the socially optimal rate, or is it not possible to tell?

PROBLEM 3.13

(Rebelo, 1991) Assume that there are two sectors, one producing consumption goods and one producing capital goods, and two factors of production: capital and land. Capital is used in both sectors, but land is used only in producing consumption goods. Specifically, the production functions are $C(t) = K_C(t)^\alpha T^{1-\alpha}$ and $\dot{K}(t) = BK_K(t)$, where K_C and K_K are the amounts of capital used in the two sectors (so $K_C(t) + K_K(t) = K(t)$) and T is the amount of land, and $0 < \alpha < 1$ and $B > 0$. Factors are paid their marginal products, and capital can move freely between the two sectors. T is normalized to 1 for simplicity.

- (a) Let $P_K(t)$ denote the price of capital goods relative to consumption goods at time t . Use the fact that the earnings of capital in units of consumption goods in the two sectors must be equal to derive a condition relating $P_K(t)$, $K_C(t)$, and the parameters α and B . If K_C is growing at rate $g_K(t)$, at what rate must P_K be growing (or falling)? Let $g_P(t)$ denote this growth rate.
- (b) The real interest rate in terms of consumption is $B + g_P(t)$.^[2] Thus, assuming that households have our standard utility function, (2.22)–(2.23), the growth rate of consumption must be $(B + g_P - \rho) / \theta \equiv g_C$. Assume $\rho < B$.
 - (i) Use your results in part (a) to express $g_C(t)$ in terms of $g_K(t)$ rather than $g_P(t)$.
 - (ii) Given the production function for consumption goods, at what rate must K_C be growing for C to be growing at rate $g_C(t)$?
 - (iii) Combine your answers to (i) and (ii) to solve for $g_K(t)$ and $g_C(t)$ in terms of the underlying parameters.
- (c) Suppose that investment income is taxed at rate τ , so that the real interest rate households face is $(1 - \tau)(B + g_P)$. How, if at all, does τ affect the equilibrium growth rate of consumption?

[2] To see this, note that capital in the investment sector produces new capital at rate B and changes in value relative to the consumption good at rate g_P . (Because the return to capital is the same in the two sectors, the same must be true of capital in the consumption sector.)

PROBLEM 3.14 *Delays in the transmission of knowledge to poor countries.*

- (a) Assume that the world consists of two regions, the North and the South. The North is described by $Y_N(t) = A_N(t)(1 - a_L)L_N$ and $\dot{A}_N(t) = a_L L_N A_N(t)$. The South does not do R&D but simply uses the technology developed in the North; however, the technology used in the South lags the North's by τ years. Thus $Y_S(t) = A_S(t)L_S$ and $A_S(t) = A_N(t - \tau)$. If the growth rate of output per worker in the North is 3 percent per year, and if a_L is close to 0, what must τ be for output per worker in the North to exceed that in the South by a factor of 10?
- (b) Suppose instead that both the North and the South are described by the Solow model: $y_i(t) = f(k_i(t))$, where

$$y_i(t) \equiv \frac{Y_i(t)}{L_i(t)} \quad \text{and} \quad k_i(t) \equiv \frac{K_i(t)}{L_i(t)}, \quad i = N, S$$

As in the Solow model, assume $\dot{K}_i(t) = sY_i(t) - \delta K_i(t)$ and $\dot{L}_i(t) = nL_i(t)$; the two countries are assumed to have the same saving rates and rates of population growth. Finally, $\dot{A}_N(t) = gA_N(t)$ and $A_S(t) = A_N(t - \tau)$.

- (i) Show that the value of k on the balanced growth path, k^* , is the same for the two countries.
- (ii) Does introducing capital change the answer to part (a)? Explain. (Continue to assume $g = 3\%$.)

PROBLEM 3.15

Which of the following possible regression results concerning the elasticity of long-run output with respect to the saving rate would provide the best evidence that differences in saving rates are not important to cross-country income differences? (1) A point estimate of 5 with a standard error of 2; (2) a point estimate of 0.1 with a standard error of 0.01; (3) a point estimate of 0.001 with a standard error of 5; (4) a point estimate of -2 with a standard error of 5. Explain your answer.

Hint: See the discussion of confidence intervals versus t -statistics in Section 3.6.

CHAPTER 4

PROBLEM 4.1 *The golden-rule level of education.*

Consider the model of Section 4.1 with the assumption that $G(E)$ takes the form $G(E) = e^{\phi E}$.

- (a) Find an expression that characterizes the value of E that maximizes the level of output per person on the balanced growth path. Are there cases where this value equals 0? Are there cases where it equals T ?
- (b) Assuming an interior solution, describe how, if at all, the golden-rule level of E (that is, the level of E you characterized in part (a)) is affected by each of the following changes:
 - (i) A rise in T .
 - (ii) A fall in n .

PROBLEM 4.2 *Endogenizing the choice of E .*

(Bils and Klenow, 2000) Suppose that the wage of a worker with education E at time t is $be^{gt}e^{\phi E}$. Consider a worker born at time 0 who will be in school for the first E years of life and will work for the remaining $T - E$ years. Assume that the interest rate is constant and equal to $\bar{r}$.

- (a) What is the present discounted value of the worker's lifetime earnings as a function of $E, T, b, \bar{r}, \phi$, and g ?
- (b) Find the first-order condition for the value of E that maximizes the expression you found in part (a). Let E^* denote this value of E . (Assume an interior solution.)
- (c) Describe how each of the following developments affects E^* :
 - (i) A rise in T .
 - (ii) A rise in $\bar{r}$.
 - (iii) A rise in g .

PROBLEM 4.3

Suppose output in country i is given by $Y_i = A_i Q_i e^{\phi E_i} L_i$. Here E_i is each worker's years of education, Q_i is the quality of education, and the rest of the notation is standard. Higher output per worker raises the quality of education. Specifically, Q_i is given by $B_i (Y_i/L_i)^\gamma$, $0 < \gamma < 1$, $B_i > 0$.

Our goal is to decompose the difference in log output per worker between two countries, 1 and 2, into the contributions of education and all other forces. We have data on Y , L , and E in the two countries, and we know the values of the parameters ϕ and γ .

- Explain in what way attributing amount $\phi(E_2 - E_1)$ of $\ln(Y_2/L_2) - \ln(Y_1/L_1)$ to education and the remainder to other forces would understate the contribution of education to the difference in log output per worker between the two countries.
- What would be a better measure of the contribution of education to the difference in log output per worker?

PROBLEM 4.4

Suppose the production function is $Y = K^\alpha (e^{\phi E} L)^{1-\alpha}$, $0 < \alpha < 1$. E is the amount of education workers receive; the rest of the notation is standard. Assume that there is perfect capital mobility. In particular, K always adjusts so that the marginal product of capital equals the world rate of return, r^* .

- Find an expression for the marginal product of capital as a function of K , E , L , and the parameters of the production function.
- Use the equation you derived in part (a) to find K as a function of r^* , E , L , and the parameters of the production function.
- Use your answer in part (b) to find an expression for $d(\ln Y)/dE$, incorporating the effect of E on Y via K .
- Explain intuitively how capital mobility affects the impact of the change in E on output.

PROBLEM 4.5

(Mankiw, Romer, and Weil, 1992) Suppose output is given by

$$Y(t) = K(t)^\alpha H(t)^\beta [A(t)L(t)]^{1-\alpha-\beta}, \quad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < 1.$$

Here L is the number of workers and H is their total amount of skills. The remainder of the notation is standard. The dynamics of the inputs are $\dot{L}(t) = nL(t)$, $\dot{A}(t) = gA(t)$, $\dot{K}(t) = s_K Y(t) - \delta K(t)$, $\dot{H}(t) = s_H Y(t) - \delta H(t)$, where $0 < s_K < 1$, $0 < s_H < 1$, and

$n + g + \delta > 0$. $L(0)$, $A(0)$, $K(0)$, and $H(0)$ are given, and are all strictly positive. Finally, define $y(t) \equiv Y(t)/[A(t)L(t)]$, $k(t) \equiv K(t)/[A(t)L(t)]$, and $h(t) \equiv H(t)/[A(t)L(t)]$.

- (a) Derive an expression for $y(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model.
- (b) Derive an expression for $\dot{k}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k, h) space, sketch the set of points where $\dot{k} = 0$.
- (c) Derive an expression for $\dot{h}(t)$ in terms of $k(t)$ and $h(t)$ and the parameters of the model. In (k, h) space, sketch the set of points where $\dot{h} = 0$.
- (d) Does the economy converge to a balanced growth path? Why or why not? If so, what is the growth rate of output per worker on the balanced growth path? If not, in general terms what is the behavior of output per worker over time?

PROBLEM 4.6

Consider the model in Problem 4.5.

- (a) What are the balanced-growth-path values of k and h in terms of s_K , s_H , and the other parameters of the model?
- (b) Suppose $\alpha = 1/3$ and $\beta = 1/2$. Consider two countries, A and B, and suppose that both s_K and s_H are twice as large in Country A as in Country B and that the countries are otherwise identical. What is the ratio of the balanced-growth-path value of income per worker in Country A to its value in Country B implied by the model?
- (c) Consider the same assumptions as in part (b). What is the ratio of the balanced-growth-path value of skills per worker in Country A to its value in Country B implied by the model?

PROBLEM 4.7

(Jones, 2002) Consider the model of Section 4.1 with the assumption that $G(E) = e^{\phi E}$. Suppose, however, that E , rather than being constant, is increasing steadily: $\dot{E}(t) = m$, where $m > 0$. Assume that, despite the steady increase in the amount of education people are getting, the growth rate of the number of workers is constant and equal to n , as in the basic model.

- (a) With this change in the model, what is the long-run growth rate of output per worker?
- (b) In the United States over the past century, if we measure E as years of schooling, $\phi \approx 0.1$ and $m \approx 1/15$. Overall growth of output per worker has been about 2

percent per year. In light of your answer to (a), approximately what fraction of this overall growth has been due to increasing education?

- (c) Can $\dot{E}(t)$ continue to equal $m > 0$ forever? Explain.

PROBLEM 4.8

Consider the following model with physical and human capital:

$$\begin{aligned} Y(t) &= [(1 - a_K) K(t)]^\alpha [(1 - a_H) H(t)]^{1-\alpha} & 0 < \alpha, a_K, a_H < 1 \\ \dot{K}(t) &= sY(t) - \delta_K K(t) \\ \dot{H}(t) &= B [a_K K(t)]^\gamma [a_H H(t)]^\phi [A(t) L(t)]^{1-\gamma-\phi} - \delta_H H(t) & \gamma, \phi > 0, \gamma + \phi < 1 \\ \dot{L}(t) &= nL(t) \\ \dot{A}(t) &= gA(t), \end{aligned}$$

where a_K and a_H are the fractions of the stocks of physical and human capital used in the education sector.

This model assumes that human capital is produced in its own sector with its own production function. Bodies (L) are useful only as something to be educated, not as an input into the production of final goods. Similarly, knowledge (A) is useful only as something that can be conveyed to students, not as a direct input to goods production.

- Define $k \equiv K/(AL)$ and $h \equiv H/(AL)$. Derive equations for $\dot{k}$ and $\dot{h}$.
- Find an equation describing the set of combinations of h and k such that $\dot{k} = 0$. Sketch in (h, k) space. Do the same for $\dot{h} = 0$.
- Does this economy have a balanced growth path? If so, is it unique? Is it stable? What are the growth rates of output per person, physical capital per person, and human capital per person on the balanced growth path?
- Suppose the economy is initially on a balanced growth path, and that there is a permanent increase in s . How does this change affect the path of output per person over time?

PROBLEM 4.9 *Increasing returns in a model with human capital.*

(Lucas, 1988) Suppose that $Y(t) = K(t)^\alpha [(1 - a_H) H(t)]^\beta$, $\dot{H}(t) = B a_H H(t)$, and $\dot{K}(t) = sY(t)$. Assume $0 < \alpha < 1$, $0 < \beta < 1$, and $\alpha + \beta > 1$.

- What is the growth rate of H ?
- Does the economy converge to a balanced growth path? If so, what are the growth rates of K and Y on the balanced growth path?

PROBLEM 4.10 *A different form of measurement error.*

Suppose the true relationship between social infrastructure (SI) and log income per person (y) is $y_i = a + bSI_i + e_i$. There are two components of social infrastructure, SI_i^A and SI_i^B (with $SI_i = SI_i^A + SI_i^B$), and we only have data on one of the components, SI_i^A . Both SI_i^A and SI_i^B are uncorrelated with e . We are considering running an OLS regression of y on a constant and SI^A .

- (a) Derive an expression of the form, $y_i = a + bSI_i^A + \text{other terms}$.
- (b) Use your answer to part (a) to determine whether an OLS regression of y on a constant and SI^A will produce an unbiased estimate of the impact of social infrastructure on income if:
 - (i) SI^A and SI^B are uncorrelated.
 - (ii) SI^A and SI^B are positively correlated.

PROBLEM 4.11

Briefly explain whether each of the following statements concerning a cross-country regression of income per person on a measure of social infrastructure is true or false:

- (a) "If the regression is estimated by ordinary least squares, it shows the effect of social infrastructure on output per person."
- (b) "If the regression is estimated by instrumental variables using variables that are not affected by social infrastructure as instruments, it shows the effect of social infrastructure on output per person."
- (c) "If the regression is estimated by ordinary least squares and has a high R^2 , this means that there are no important influences on output per person that are omitted from the regression; thus in this case, the coefficient estimate from the regression is likely to be close to the true effect of social infrastructure on output per person."

PROBLEM 4.12 *Convergence regressions.*

- (a) **Convergence.** Let y_i denote log output per worker in country i . Suppose all countries have the same balanced-growth-path level of log income per worker, y^* . Suppose also that y_i evolves according to $dy_i(t)/dt = -\lambda [y_i(t) - y^*]$.
 - (i) What is $y_i(t)$ as a function of $y_i(0)$, y^* , λ , and t ?
 - (ii) Suppose that $y_i(t)$ in fact equals the expression you derived in part (i) plus a mean-zero random disturbance that is uncorrelated with $y_i(0)$. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$.

What is the relation between β , the coefficient on $y_i(0)$ in the regression, and λ , the speed of convergence? Given this, how could you estimate λ from an estimate of β ?

Hint: For a univariate OLS regression, the coefficient on the right-hand-side variable equals the covariance between the right-hand-side and left-hand-side variables divided by the variance of the right-hand-side variable.

- (iii) If β in part (ii) is negative (so that rich countries on average grow less than poor countries), is $\text{Var}(y_i(t))$ necessarily less than $\text{Var}(y_i(0))$, so that the cross-country variance of income is falling? Explain. If β is positive, is $\text{Var}(y_i(t))$ necessarily more than $\text{Var}(y_i(0))$? Explain.

- (b) **Conditional convergence.** Suppose $y_i^* = a + bX_i$, and that

$$\frac{dy_i(t)}{dt} = -\lambda [y_i(t) - y_i^*]$$

- (i) What is $y_i(t)$ as a function of $y_i(0)$, X_i , λ , and t ?
- (ii) Suppose that $y_i(0) = y_i^* + u_i$ and that $y_i(t)$ equals the expression you derived in part (i) plus a mean-zero random disturbance, e_i , where X_i , u_i , and e_i are uncorrelated with one another. Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \varepsilon_i$. Suppose one attempts to infer λ from the estimate of β using the formula in part (a)(ii). Will this lead to a correct estimate of λ , an overestimate, or an underestimate?
- (iii) Consider a cross-country growth regression of the form $y_i(t) - y_i(0) = \alpha + \beta y_i(0) + \gamma X_i + \varepsilon_i$. Under the same assumptions as in part (ii), how could one estimate b , the effect of X on the balanced-growth-path value of y , from estimates of β and γ ?

CHAPTER 5

PROBLEM 5.1

Redo the calculations reported in Table 5.1, 5.2, or 5.3 for any country other than the United States.

PROBLEM 5.2

Redo the calculations reported in Table 5.3 for the following:

- (a) Employees' compensation as a share of national income.
- (b) The labor force participation rate.
- (c) The federal government budget deficit as a share of GDP.
- (d) The Standard and Poor's 500 composite stock price index.
- (e) The difference in yields between Moody's Baa and Aaa bonds.
- (f) The difference in yields between 10-year and 3-month U.S. Treasury securities.
- (g) The weighted average exchange rate of the U.S. dollar against major currencies.

PROBLEM 5.3

Let A_0 denote the value of A in period 0, and let the behavior of $\ln A$ be given by equations (5.8)–(5.9).

- (a) Express $\ln A_1$, $\ln A_2$, and $\ln A_3$ in terms of $\ln A_0$, ε_{A1} , ε_{A2} , ε_{A3} , $\bar{A}$, and g .
- (b) In light of the fact that the expectations of the ε_A 's are zero, what are the expectations of $\ln A_1$, $\ln A_2$, and $\ln A_3$ given $\ln A_0$, $\bar{A}$, and g ?

PROBLEM 5.4

Suppose the period- t utility function, u_t , is $u_t = \ln c_t + b(1 - \ell_t)^{1-\gamma} / (1 - \gamma)$, $b > 0$, $\gamma > 0$, rather than (5.7).

- (a) Consider the one-period problem analogous to that investigated in (5.12)–(5.15). How, if at all, does labor supply depend on the wage?

- (b) Consider the two-period problem analogous to that investigated in (5.16)–(5.21). How does the relative demand for leisure in the two periods depend on the relative wage? How does it depend on the interest rate? Explain intuitively why γ affects the responsiveness of labor supply to wages and the interest rate.

PROBLEM 5.5

Consider the problem investigated in (5.16)–(5.21).

- (a) Show that an increase in both w_1 and w_2 that leaves w_1/w_2 unchanged does not affect ℓ_1 or ℓ_2 .
- (b) Now assume that the household has initial wealth of amount $Z > 0$.
- (i) Does (5.23) continue to hold? Why or why not?
- (ii) Does the result in (a) continue to hold? Why or why not?

PROBLEM 5.6

Suppose an individual lives for two periods and has utility $\ln C_1 + \ln C_2$.

- (a) Suppose the individual has labor income of Y_1 in the first period of life and zero in the second period. Second-period consumption is thus $(1 + r)(Y_1 - C_1)$; r , the rate of return, is potentially random.
- (i) Find the first-order condition for the individual's choice of C_1 .
- (ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?
- (b) Suppose the individual has labor income of zero in the first period and Y_2 in the second. Second-period consumption is thus $Y_2 - (1 + r)C_1$. Y_2 is certain; again, r may be random.
- (i) Find the first-order condition for the individual's choice of C_1 .
- (ii) Suppose r changes from being certain to being uncertain, without any change in $E[r]$. How, if at all, does C_1 respond to this change?

PROBLEM 5.7

- (a) Use an argument analogous to that used to derive equation (5.23) to show that

household optimization requires

$$\frac{b}{1 - \ell_t} = e^{-\rho} E_t \left[\frac{b w_t (1 + r_{t+1})}{w_{t+1} (1 - \ell_{t+1})} \right]$$

- (b) Show that this condition is implied by (5.23) and (5.26). (Note that [5.26] must hold in every period.)

PROBLEM 5.8 *A simplified real-business-cycle model with additive technology shocks.*

(Blanchard and Fischer, 1989, pp. 329–331) Consider an economy consisting of a constant population of infinitely lived individuals. The representative individual maximizes the expected value of $\sum_{t=0}^{\infty} u(C_t) / (1 + \rho)^t$, $\rho > 0$. The instantaneous utility function, $u(C_t)$, is $u(C_t) = C_t - \theta C_t^2$, $\theta > 0$. Assume that C is always in the range where $u'(C)$ is positive.

Output is linear in capital, plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation; thus $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A . Assume $A = \rho$. Finally, the disturbance follows a first-order autoregressive process: $e_t = \phi e_{t-1} + \varepsilon_t$, where $-1 < \phi < 1$ and where the ε_t 's are mean-zero, i.i.d. shocks.

- Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1} .
- Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma e_t$. Given this guess, what is K_{t+1} as a function of K_t and e_t ?
- What values must the parameters α , β , and γ have for the first-order condition in part (a) to be satisfied for all values of K_t and e_t ?
- What are the effects of a one-time shock to ε on the paths of Y , K , and C ?

PROBLEM 5.9 *A simplified real-business-cycle model with taste shocks.*

(Blanchard and Fischer, 1989, p. 361) Consider the setup in Problem 5.8. Assume, however, that the technological disturbances (the e 's) are absent and that the instantaneous utility function is $u(C_t) = C_t - \theta (C_t + v_t)^2$. The v 's are mean-zero, i.i.d. shocks.

- Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1} .
- Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma v_t$. Given this guess, what is K_{t+1} as a function of K_t and v_t ?
- What values must the parameters α , β , and γ have for the first-order condition in (a) to be satisfied for all values of K_t and v_t ?
- What are the effects of a one-time shock to v on the paths of Y , K , and C ?

PROBLEM 5.10 *The balanced growth path of the model of Section 5.3.*

Consider the model of Section 5.3 without any shocks. Let y^* , k^* , c^* , and G^* denote the values of $Y/(AL)$, $K/(AL)$, $C/(AL)$, and $G/(AL)$ on the balanced growth path; w^* the value of w/A ; ℓ^* the value of L/N ; and r^* the value of r .

- (a) Use equations (5.1)–(5.4), (5.23), and (5.26) and the fact that y^* , k^* , c^* , w^* , ℓ^* , and r^* are constant on the balanced growth path to find six equations in these six variables.

Hint: The fact that c in [5.23] is consumption per person, C/N , and c^* is the balanced-growth-path value of consumption per unit of effective labor, $C/(AL)$, implies that $c = c^* \ell^* A$ on the balanced growth path.

- (b) Consider the parameter values assumed in Section 5.7. What are the implied shares of consumption and investment in output on the balanced growth path? What is the implied ratio of capital to annual output on the balanced growth path?

PROBLEM 5.11 *Solving a real-business-cycle model by finding the social optimum.*

Consider the model of Section 5.5. Assume for simplicity that $n = g = \bar{A} = \bar{N} = 0$. Let $V(K_t, A_t)$, the *value function*, be the expected present value from the current period forward of lifetime utility of the representative individual as a function of the capital stock and technology.

- (a) Explain intuitively why $V(\cdot)$ must satisfy

$$V(K_t, A_t) = \max_{C_t, \ell_t} \{[\ln C_t + b \ln(1 - \ell_t)] + e^{-\rho} E_t [V(K_{t+1}, A_{t+1})]\}.$$

This condition is known as the *Bellman equation*.

Given the log-linear structure of the model, let us guess that $V(\cdot)$ takes the form $V(K_t, A_t) = \beta_0 + \beta_K \ln K_t + \beta_A \ln A_t$, where the values of the β 's are to be determined. Substituting this conjectured form and the facts that $K_{t+1} = Y_t - C_t$ and $E_t[\ln A_{t+1}] = \rho_A \ln A_t$ into the Bellman equation yields

$$V(K_t, A_t) = \max_{C_t, \ell_t} \{[\ln C_t + b \ln(1 - \ell_t)] + e^{-\rho} [\beta_0 + \beta_K \ln(Y_t - C_t) + \beta_A \rho_A \ln A_t]\}.$$

- (b) Find the first-order condition for C_t . Show that it implies that C_t/Y_t does not depend on K_t or A_t .
- (c) Find the first-order condition for ℓ_t . Use this condition and the result in part (b) to show that ℓ_t does not depend on K_t or A_t .
- (d) Substitute the production function and the results in parts (b) and (c) for the op-

timal C_t and ℓ_t into the equation above for $V(\cdot)$, and show that the resulting expression has the form $V(K_t, A_t) = \beta'_0 + \beta'_K \ln K_t + \beta'_A \ln A_t$.

- (e) What must β_K and β_A be so that $\beta'_K = \beta_K$ and $\beta'_A = \beta_A$?
- (f) What are the implied values of C/Y and ℓ ? Are they the same as those found in Section 5.5 for the case of $n = g = 0$?

PROBLEM 5.12

Suppose technology follows some process other than (5.8)–(5.9). Do $s_t = \hat{s}$ and $\ell_t = \hat{\ell}$ for all t continue to solve the model of Section 5.5? Why or why not?

PROBLEM 5.13

Consider the model of Section 5.5. Suppose, however, that the instantaneous utility function, u_t , is given by $u_t = \ln c_t + b(1 - \ell_t)^{1-\gamma} / (1 - \gamma)$, $b > 0$, $\gamma > 0$, rather than by (5.7) (see Problem 5.4).

- (a) Find the first-order condition analogous to equation (5.26) that relates current leisure and consumption, given the wage.
- (b) With this change in the model, is the saving rate (s) still constant?
- (c) Is leisure per person ($1 - \ell$) still constant?

PROBLEM 5.14

- (a) If the $\tilde{A}_t$'s are uniformly 0 and if $\ln Y_t$ evolves according to (5.39), what path does $\ln Y_t$ settle down to?

Hint: Note that we can rewrite [5.39] as

$$\ln Y_t - (n + g)t = Q + \alpha [\ln Y_{t-1} - (n + g)(t - 1)] + (1 - \alpha)\tilde{A}_t$$

where $Q \equiv \alpha \ln \hat{s} + (1 - \alpha)(\bar{A} + \ln \hat{\ell} + \bar{N}) - \alpha(n + g)$.

- (b) Let $\tilde{Y}_t$ denote the difference between $\ln Y_t$ and the path found in (a). With this definition, derive (5.40).

PROBLEM 5.15 *The derivation of the log-linearized equation of motion for capital.*

Consider the equation of motion for capital, $K_{t+1} = K_t + K_t^\alpha (A_t L_t)^{1-\alpha} - C_t - G_t - \delta K_t$.

(a) (i) Show that

$$\frac{\partial \ln K_{t+1}}{\partial \ln K_t} = \frac{1 + r_t}{K_{t+1}/K_t}$$

(ii) Show that this implies that $\partial \ln K_{t+1} / \partial \ln K_t$ evaluated at the balanced growth path is $(1 + r^*) / e^{n+g}$.

(b) Show that

$$\tilde{K}_{t+1} \simeq \lambda_1 \tilde{K}_t + \lambda_2 (\tilde{A}_t + \tilde{L}_t) + \lambda_3 \tilde{G}_t + (1 - \lambda_1 - \lambda_2 - \lambda_3) \tilde{C}_t,$$

where

$$\lambda_1 \equiv \frac{(1 + r^*)}{e^{n+g}}, \quad \lambda_2 \equiv \frac{(1 - \alpha)(r^* + \delta)}{\alpha e^{n+g}}, \quad \lambda_3 \equiv -\frac{(r^* + \delta)(G/Y)^*}{\alpha e^{n+g}}$$

and where $(G/Y)^*$ denotes the ratio of G to Y on the balanced growth path without shocks. (Hints: Since the production function is Cobb–Douglas, $Y^* = (r^* + \delta) K^* / \alpha$. On the balanced growth path, $K_{t+1} = e^{n+g} K_t$, which implies that $C^* = Y^* - G^* - \delta K^* - (e^{n+g} - 1) K^*$.)

(c) Use the result in (b) and equations (5.43)–(5.44) to derive (5.53), where $b_{KK} = \lambda_1 + \lambda_2 a_{LK} + (1 - \lambda_1 - \lambda_2 - \lambda_3) a_{CK}$, $b_{KA} = \lambda_2 (1 + a_{LA}) + (1 - \lambda_1 - \lambda_2 - \lambda_3) a_{CA}$, and $b_{KG} = \lambda_2 a_{LG} + \lambda_3 + (1 - \lambda_1 - \lambda_2 - \lambda_3) a_{CG}$.

PROBLEM 5.16

Redo the regression reported in equation (5.55):

- (a) Incorporating more recent data.
- (b) Incorporating more recent data, and using $M1$ rather than $M2$.
- (c) Including eight lags of the change in log money rather than four.

REFERENCES

A

Allais, Maurice. 1947. *Économie et Intérêt*. Paris: Imprimerie Nationale.

B

Barro, Robert J., N. Gregory Mankiw, and Xavier Sala-i-Martin. 1995. "Capital Mobility in Neoclassical Models of Growth." *American Economic Review* 85 (March): 103–115.

Bils, Mark J. and Peter J. Klenow. 2000. "Does Schooling Cause Growth?" *American Economic Review* 90 (December): 1160–1183.

Black, Fischer. 1974. "Uniqueness of the Price Level in Monetary Growth Models with Rational Expectations." *Journal of Economic Theory* 7 (January): 53–65.

Blanchard, Olivier J. and Stanley Fischer. 1989. *Lectures on Macroeconomics*. Cambridge, MA: MIT Press.

Brock, William. 1975. "A Simple Perfect Foresight Monetary Model." *Journal of Monetary Economics* 1 (April): 133–150.

C

Calvo, Guillermo. 1978. "On the Indeterminacy of Interest Rates and Wages with Perfect Foresight." *Journal of Economic Theory* 19 (December): 321–337.

J

Jones, Charles I. 1995. "R&D-Based Models of Economic Growth." *Journal of Political Economy* 103 (August): 759–784.

Jones, Charles I. 2002. "Sources of U.S. Economic Growth in a World of Ideas." *American Economic Review* 92 (March): 220–239.

L

Lucas, Robert E., Jr. 1988. "On the Mechanics of Economic Development." *Journal of Monetary Economics* 22 (July): 3–42.

M

Mankiw, N. Gregory, David Romer, and David N. Weil. 1992. "A Contribution to the Empirics of Economic Growth." *Quarterly Journal of Economics* 107 (May): 407–437.

P

Piketty, Thomas. 2014. *Capital in the Twenty-First Century*. Trans. by Arthur Goldhammer. Cambridge, MA: Harvard University Press.

Piketty, Thomas and Gabriel Zucman. 2014. "Capital Is Back: Wealth-Income Ratios in Rich Countries 1700–2010." *Quarterly Journal of Economics* 129 (August): 1255–1310.

R

Rebelo, Sergio. 1991. "Long-Run Policy Analysis and Long-Run Growth." *Journal of Political Economy* 99 (June): 500–521.

Rognlie, Matthew. 2015. "Deciphering the Fall and Rise in the Net Capital Share: Accumulation or Scarcity?" *Brookings Papers on Economic Activity*, no. 1: 1–54.

S

Samuelson, Paul A. 1958. "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money." *Journal of Political Economy* 66 (December): 467–482.

Sato, K. 1966. "On the Adjustment Time in Neo-Classical Growth Models." *Review of Economic Studies* 33 (July): 263–268.

Shell, Karl. 1971. "Notes on the Economics of Infinity." *Journal of Political Economy* 79 (September): 1002–1011.

Smith, William T. 2006. "A Closed Form Solution to the Ramsey Model." *Contributions to Macroeconomics* 6:1, Article 3.

Solow, Robert M. 1960. "Investment and Technical Progress." In: *Mathematical Methods in the Social Sciences 1959*. Ed. by Kenneth J. Arrow, Samuel Karlin, and Patrick Suppes. Stanford: Stanford University Press, pp. 89–104.